

AGRO STARTUP

Design and Fabrication of Electric Cultivator

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ABSTRACT

The Electric Cultivator Project proposes the design and development of an electric cultivator equipped with a rotavator via PTO shaft, targeting efficient, eco-friendly tillage solutions for small to medium scale farming. The traditional petrol – powered cultivators is widely used but it poses multiple challenges such as high emission of pollutants, high fuel and maintenance costs, noise pollution, and can be heavy and difficult to operate for extended periods. These drawbacks make them less accessible and sustainable, especially for smallholder farmers seeking cost-effective and environmentally responsible tools.

The electric cultivator seeks to address these issues by incorporating a high torque motor such as a Permanent Magnet Synchronous motor (PMSM) or a Brushless DC motor (BLDC), which can deliver performance equivalent to petrol engines. The electric motor reduces emission and noise, lowering the environmental impact and improving usability. Furthermore, the electric cultivator minimizes maintenance needs, as it does not require fuel or oil changes, and offers a lightweight, user – friendly design that reduces physical strain for operators.

This project will encompass comprehensive design and prototyping, with a focus on optimizing the torque and efficiency of the electric rotavator for varied soil conditions. Field testing will be conducted to ensure robust performance and assess operational benefits, including cost savings, ease of use and sustainability. The proposed electric cultivator not only enhances productivity but also aligns with goals for sustainable agriculture, providing an innovative alternative to traditional tillage tools.

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LIST OF ABBREVIATIONS AND ACRONYMS

PMSM: Permanent Magnet Synchronous Motor

BLDC; Brushless Direct Current

Chapter 1: Introduction:

1.1 Background

Agricultural mechanisation has made remarkable progress in recent years, revolutionising traditional farming methods with more efficient, reliable and eco – friendly alternatives. In this context, the electric cultivator represents a significant advancement over traditional petrol-based cultivators, addressing both environmental and operational limitations. This project proposes the development of an electric cultivator equipped with a Permanent Magnet Synchronous Motor (PMSM) for providing motion to the cultivator and Brushless DC Motor (BLDC) for rotating the rotavator, each capable of delivering sufficient torque for effective tillage operation while offering several advantages over conventional petrol – powered cultivators.

Electric cultivators are increasingly favoured for their reduced emissions, quieter operation, and lower maintenance requirements. By eliminating the need for petrol, they offer a cleaner, greener alternative, aligning with global efforts to reduce carbon emissions and dependence on fossil fuels. Furthermore, the enhanced torque capabilities of PMSM and BLDC can support deeper tillage and more effective soil preparation, making them suitable for a wide range of agricultural tasks. These motor types also offer higher efficiency and precise control, optimising power usage and reducing overall operational costs.

The objective of this project is to design an electric cultivator prototype tailored to small – scale farming needs, where manoeuvrability, cost-effectiveness, and ease of operation are essential. The proposed electric cultivator will be designed to handle various soil types, offering flexibility for small – scale farmers. By employing advanced motor technologies and innovative design, the project aims to demonstrate how electric cultivators can enhance agricultural productivity while promoting sustainable practices.

The proposal outlines the design, development, and testing phases necessary to realise an efficient, eco-friendly electric cultivator. It also highlights the expected benefits in terms of reduced emissions, operational efficiency and cost-savings, ultimately contributing to more sustainable and modern farming solutions.

1.2 Problem Statement

Demand for ecologically suitable alternatives for conventional gasoline-powered farm equipment has increased as a result of the increased focus on sustainable agriculture and lower carbon emissions worldwide. Despite their effectiveness, traditional gasoline-powered cultivators have a substantial impact on greenhouse gas emissions, noise pollution, and operating expenses. Because they depend on fossil fuels, their use is not only bad for the environment but also sensitive to changes in fuel prices, which has an impact on the whole cost of farming operations. Therefore, it is essential to create cost-effective and sustainable solutions for small and medium-sized farmers, especially in developing nations where farming is both economically vital and financially limited.

Given their potential for cheaper operating costs, lower noise levels, and zero exhaust emissions, electric cultivators provide an appealing substitute. However, there are unique obstacles in the way of the development of electric cultivators. First and foremost, the torque and power output specifications required for efficient tillage must be continuously fulfilled. This involves choosing between many electric motor technologies, each with unique performance traits, efficiencies, and financial implications such as Brushless DC (BLDC) motors and Permanent Magnet Synchronous Motors (PMSM).

Second, in order to minimize field downtime, the energy source and storage for electric cultivators—typically lithium-ion batteries—must be adjusted for long-lasting performance. Particularly in rural or off-grid areas, battery life, charging infrastructure, and overall energy management are important variables that affect the usability and adoption of these electric alternatives.

Furthermore, because electric motors typically produce less torque than their gasoline-powered counterparts, building an electric cultivator necessitates a detailed understanding of the load needs for tillage. For electric cultivators to meet the performance requirements of conventional equipment, these torque limitations must be addressed without sacrificing durability or efficiency.

Last but not least, affordability is crucial because small-scale farmers want dependable yet reasonably priced equipment. The design of electric cultivators must strike a balance between cost, performance, and ease of maintenance in order to make these devices affordable and appealing to a wider range of users. Therefore, creative design solutions are needed to address

the particular constraints presented by agricultural applications in order to build an electric cultivator that is competitive, efficient, and environmentally sustainable.

1.3 OBJECTIVES

1.3.1 Primary Objective:

- a) **Create an Eco – friendly Cultivator:** Create an emission free electric cultivator that provides a sustainable substitute for conventional gasoline – powered equipment.
- b) **Achieve Adequate Torque and Power Output:** Make sure the electric motors (BLDC and PMSM) provide enough torque and power requirements for efficient tillage.
- c) **Increase Battery Life and Energy Efficiency:** Create an effective energy management and storage system with an emphasis on extending field usage by reducing downtime and increasing battery life.
- d) **Assure Cost – Effectiveness for Small – Scale Farmers:** Create an electric cultivator that is both economical and inexpensive that offers small and medium sized farmers a low – cost option that satisfies their budgetary requirements.

1.3.2 Secondary Objective:

- a) **Minimize Noise and Vibration Levels:** Design the cultivator at lower noise and vibration levels compared to petrol models, enhancing comfort and usability.
- b) **Simplify Maintenance and Repair Requirements:** Develop a low – maintenance design with easily replaceable parts to reduce operational downtime and repair costs.
- c) **Integrate Smart Features for Improved Usability:** Explore the potential for smart technology integration, such as battery status indicators, torque sensors, to improve user experience and optimize performance.
- d) **Enhance Durability for Rough Agricultural Conditions:** Ensure that the cultivator's build quality and material can withstand tough farming environments, maximizing the lifespan of the equipment.

1.4 Significance

1) Promoting Sustainable Agriculture

- Supports eco-friendly farming by reducing greenhouse gas emissions and air pollution.
- Aligns with global sustainability goals by utilising clean energy alternatives.

2) Enhanced Efficiency and Reduced Operational Costs:

- Lowers long – term costs by eliminating fuel expenses and leveraging energy – efficient technology.
- Provides cost savings to small-scale farmers, making farming operations more affordable.

3) Improved Soil and Operator Health:

- Reduces soil contamination from exhaust emissions, helping maintain soil quality.
- Decreases noise levels, creating a safer and more comfortable working environment.

4) Meeting the needs of small – scale farmers:

- Offers an affordable. Compact solution suited to small farmers and diverse field conditions.
- Fills a gap in the market for small – scale, affordable agricultural machinery.

5) Technological Advancements in Agricultural Machinery:

- Contributes to innovation in electric motor design, energy storage and smart control systems.
- Supports the broader movement towards automation and smart farming practices.

6) Reducing Dependence on Fossil Fuels

- Decreases reliance on petrol, promoting stability against fluctuating fuel prices.
- Lowers the environmental footprint of farming operations.

7) Economic Opportunities and Skill development:

- Creates job opportunities in manufacturing, maintenance, sales of electric machinery.
- Encourages skill development in electric motor maintenance, battery management and electronics.

1.5 Market of Cultivators in Nepal:

The agricultural sector is the backbone of Nepal's economy, employing a majority of the population. Cultivators, including traditional and modern variants, play a vital role in enhancing productivity by preparing the soil efficiently for farming. Nepal's market for

cultivators has been steadily growing, driven by increasing mechanisation in agriculture, government subsidies, and the demand for efficient farming tools in rural areas. Small-scale and compact cultivators are particularly popular, catering to the needs of fragmented and terraced farmlands typical in the country. Imports dominate the market, with machines from India, China, and Japan being the most common. However, there is significant potential for locally manufactured cultivators to reduce dependence on imports and address specific requirements, such as rugged terrain and affordability.

The Key market factors are:

- i. Dependence on Agriculture
 - Agriculture is a primary sector in Nepal, employing the majority of the population.
 - Cultivators are essential for improving farming efficiency and soil preparation.
- ii. Rising Mechanisation
 - Labour shortages and the need for higher productivity are driving increased adoption of mechanised tools like cultivators.
- iii. Government Support
 - Subsidies and grants under agricultural modernisation programmes make cultivators more accessible to small and medium-scale farmers.
- iv. Terrain-Specific Needs
 - Nepal's fragmented and terraced farmlands favour small-scale, lightweight, and compact cultivators.
- v. Market Composition
 - Imports dominate, with cultivators primarily sourced from India, China, and Japan.
 - Limited local manufacturing creates a reliance on foreign products.
- vi. Opportunities for Local Manufacturing
 - Demand for affordable and terrain-specific machines highlights the potential for domestic production.
- vii. Consumer Preferences
 - Ease of maintenance, fuel efficiency, and adaptability to different soil conditions significantly influence buying decisions.
- viii. Sustainability Trends
 - The growing interest in electric and hybrid cultivators aligns with global trends in sustainable agriculture, offering new market opportunities.

1.6 Market Analysis:

The cultivator market in Nepal is diverse, with strong competition from imported brands and emerging local manufacturers. Understanding the competitive landscape is crucial for identifying market opportunities and challenges. The competition primarily revolves around price, quality, adaptability to Nepal's terrain, and after-sales support. This analysis highlights the strengths, weaknesses, and positioning of key players in the market to provide insights into their strategies and market dynamics.

Key Competitors in the Nepalese Cultivator Market

1. Conventional Petrol/Diesel Cultivators

❖ Strengths:

- Widely available in Nepal.
- Established supply chains and servicing networks.
- Higher torque suitable for hard soils.
- Weaknesses:
- High operational costs due to rising fuel prices.
- Noise, maintenance, and environmental concerns.
- Unsuitability for small-scale fields due to bulkiness.

2. Imported Electric Cultivators

❖ Strengths:

- International brands (e.g., Honda, Husqvarna) offer quality and reliability.
- Some models cater to niche, environmentally conscious farmers.
- Weaknesses:
- High initial costs due to import duties and shipping.
- Limited after-sales service and spare parts availability in rural Nepal.
- Unsuitability for the specific needs of Nepalese terrain and farming practices.

3. Local Manufacturers of Agricultural Equipment

❖ Strengths:

- Cost-effective solutions tailored to the Nepalese market.
- Easier access to maintenance services and spare parts.

❖ Weaknesses:

- Lack of innovation in electric cultivator design.
- Limited capacity to produce high-quality electric machinery.

Market Positioning of Competitors

- Market Leaders: Indian brands due to their affordability and adaptability to regional needs.
- Niche Players: Japanese brands targeting premium segments requiring advanced technology.
- Emerging Competitors: Local manufacturers gaining traction by addressing specific terrain needs and offering affordable solutions.

Competitive Advantages for Local Manufacturing

- Customisation for Nepal's unique topography and soil conditions.
- Cost-effective solutions due to reduced import tariffs and local sourcing.
- Opportunities to integrate sustainable technologies, such as electric and hybrid cultivators.
- Strategic Implications

To remain competitive, local manufacturers and new entrants need to focus on affordability, terrain-specific designs, and after-sales support while leveraging advancements in sustainable technologies.

Chapter 2. Literature Review

As the demand for sustainable and low maintenance agricultural solution grows, the market for electric cultivators has experienced significant expansion. Electric cultivators are preferred among famers, urban gardeners and environmentally-conscious growers due to their eco – friendly operation and lower noise output compared to traditional petrol – powered models. The literature review below explores five key aspect of market available electric cultivators: adoption trends, advancement in battery technology, motor efficiency, advantages and limitations and consumer preferences.

1. Growth and Adoption of Electric Cultivators in Agriculture [1]:

The market for electric farming equipment has been increased significantly by the global shift towards sustainable farming practices. Small scale farmers are increasingly adapting towards electrical equipment as a sustainable alternative towards petrol – powered models. According to **Hossain et al. (2021)**, the adoption rate for electric cultivators are rising, primarily due to their lower operational costs, reduced maintenance, eco-friendly attributes. Electric cultivators have demonstrated effectiveness in small farms and urban gardening settings where large petrol-powered models are not feasible. Hossain’s study emphasizes that electric cultivators re best suited for light moderate tillage tasks, with many models targeted towards home gardeners or small agricultural operations rather than commercial or heavy-duty farming.

This study also highlights that the demand for electric cultivators is expected to grow significantly in the coming years due to the advancements in motor technology and battery efficiency. Regions with high rated of urban agriculture, particularly in Europe and parts of Asia have witnessed accelerated adoptions. For example, in Germany and Japan, government initiatives aimed at reducing carbon emissions in agriculture have further incentivised the use of electric cultivators. **Hossain et al.** predic.t that these machines continue to evolve, they could become more competitive with petrol cultivators, even in more challenging agricultural tasks.

2. Advancement in Battery technology [2]:

One important element affecting the use and effectiveness of electric cultivators is battery performance. Heavy, ineffective batteries frequently hampered the operation duration and overall feasibility of early electric cultivators. However, the functionality and performance of

these devices have been greatly enhanced by recent developments in battery technology, especially with regard to lithium-ion batteries. According to Zhang and Liu's (2022) comprehensive analysis of battery technology in agricultural applications, lithium-ion batteries are now the norm in the majority of contemporary electric cultivators because of their high energy density, low weight, and quick charging speed.

However, the study also identifies issues with battery cost and longevity: the high initial cost of lithium-ion batteries can raise the purchase price of electric cultivators, which may deter some consumers; additionally, battery degradation over time reduces the efficiency and lifespan of these machines, especially if used frequently in heavy-duty applications; and finally, Zhang and Liu suggest that future advancements in battery technology, such as the integration of solid-state batteries, could further enhance the viability of electric cultivators, providing even longer runtimes and faster charging capabilities.

3. Motor types and Efficiency in Electric Cultivators [3]:

Permanent Magnet Synchronous Motors (PMSM) and Brushless Direct Current (BLDC) motors are the two main motor types used in electric cultivators. Every motor type has unique benefits and is chosen according to the cultivator's particular needs. In their comparison of BLDC and PMSM motors in the context of electric cultivators, Jain and Kumar (2020) pointed out that BLDC motors are frequently found in lightweight cultivators because of their inexpensive cost, simple control systems, and good efficiency at low power levels. Because of these features, BLDC motors are perfect for home gardening and small-scale agricultural practices.

On the other hand, PMSM motors are better appropriate for heavy-duty electric cultivators because they provide more torque density and efficiency, particularly at higher power levels. In hard or clay-rich soils, where more torque is necessary for efficient plowing, Jain and Kumar discovered that cultivators with PMSM motors work better. But the report also points out that PMSM motors are often more costly, which can raise the price of cultivators made for heavier-duty uses. According to the authors, choosing a motor should take into account the kind of soil and the intensity of the task; BLDC motors are best suited for light- to medium-duty applications, while PMSM motors are better for heavier-duty requirements.

4. Advantages and Limitations of Electric Cultivators [4]

According to Evans and Brown's (2019) thorough analysis, electric cultivators have a number of advantages over their petrol-powered counterparts. For example, they are quieter, require less maintenance, and emit no direct emissions, making them perfect for urban agriculture where noise restrictions and emission concerns are common. Additionally, because they don't have the complicated engine components that petrol models do, electric cultivators are generally easier to operate and maintain, which lowers operating costs and the need for technical repairs.

Evans and Brown do point out some restrictions, though. Even with advancements in battery and motor technology, electric cultivators frequently lack the raw power required for demanding, extensive tillage operations. Specifically, extensive tilling on heavy or rocky soil may be difficult for models made for small and medium farms. Another drawback is battery life; although many growers now provide lithium-ion batteries with respectable run periods, these batteries can still deplete rapidly under demanding conditions, necessitating frequent recharging or the usage of numerous batteries. Because of this, electric cultivators are now more appropriate for usage in small-scale or urban farming settings and in lighter soil types than in commercial agriculture.

5. Consumer Preferences and Market Penetration [5]

The market growth of electric cultivators is significantly influenced by consumer preferences. In a survey of home growers, urban gardeners, and small-scale farms, Williams et al. (2023) discovered that the main reasons why consumers chose electric cultivators over gasoline-powered ones are ease of maintenance and environmental advantages. According to their research, consumers value electric cultivators' quieter operation and zero emissions, which make them appropriate for densely populated locations. Additionally, non-professional users who do not have the technical know-how to maintain petrol engines are drawn to the lesser maintenance requirements.

The study also discovered that cost is still a major barrier to wider use in spite of these advantages. Some buyers, especially those in underdeveloped nations, may be put off by the greater initial cost of electric cultivators, especially those with sophisticated battery and motor technology. Some prospective consumers are also deterred by issues with battery lifespan and the requirement for regular recharging. According to Williams et al., raising consumer

knowledge of electric cultivators' long-term financial and environmental advantages may help them get beyond these obstacles and increase their market share.

Conclusion:

With an emphasis on their benefits, drawbacks, and market variables affecting consumer decisions, this literature review provides an overview of the current status of electric cultivators in the agriculture industry. Despite their growing popularity among urban gardeners and small-scale farmers, electric cultivators' power and battery life constraints limit their use for larger-scale uses. Although cost is still a limiting factor, recent developments in battery technology, such as the creation of lithium-ion and possibly solid-state batteries, have improved their performance.

Electric cultivators may become more competitive as a result of future advancements in battery and motor technology, making them a competitive option even for more demanding agricultural jobs. As developments overcome current constraints, the sustained interest in sustainable farming methods points to a bright future for electric cultivators. According to market trends, electric cultivators could become a vital component of present-day, sustainable agricultural methods as consumer awareness and technology advance.

Chapter 3. Methodology

The methodology for completing the electric cultivator project outlines the systematic approach and steps taken to ensure efficient development, testing, and validation. Here's a breakdown of the methodology:

Define Project Scope and Objectives

- Outline key objectives, including power, efficiency, and cost targets.
- Establish desired cultivator performance metrics (e.g., tilling depth, speed, and width).

Motor Selection and Analysis

- Compare PMSM and BLDC motors in terms of torque, efficiency, and control complexity.

Power Supply and Battery Sizing

- Determine the required battery capacity based on motor power and operating time.
- Choose a suitable battery chemistry (e.g., Li-ion for higher energy density and lifespan).
- Design a power management system to optimise power delivery and conserve battery life.

Electrical System Design

- Design a control system for efficient motor operation and speed control.
- Implement safety measures (e.g., circuit protection and emergency shutdown).
- Integrate sensors for feedback (e.g., RPM and load sensors) for real-time monitoring.

Mechanical Structure and Transmission Design

- Design the transmission system, considering gear ratios for torque multiplication.

Blade and Rotavator Assembly Design

- Calculate blade geometry and spacing to maximise tilling efficiency.
- Select appropriate materials and treatments for blade durability.
- Design mounting and attachment points to enable easy blade replacement.

Prototype Development and Testing

- Build a prototype for initial testing and validation of design assumptions.
- Conduct field tests to assess tilling performance, battery life, and system efficiency.

Optimization and Iteration

- Refine motor control algorithms for better efficiency and responsiveness.
- Adjust mechanical and electrical components based on test results.
- Improve ergonomics and usability features for end users.

Final Testing and Validation

- Perform comprehensive field testing under varying soil and operational conditions.
- Validate safety, reliability, and durability over extended use.

Documentation and Compliance

- Document design choices, calculations, and test data.
- Ensure the design meets regulatory and safety standards for electric cultivators.

Chapter 4. Components of the Electric Cultivator:

4.1 Motor:

A **motor** is an electrical device that converts electrical energy into mechanical motion, allowing machines to perform various tasks. In the case of the electric cultivator, the motor provides the necessary rotational power to drive the cultivator's components such as the rotavator blades, wheels, and differential. The motor is a critical component for powering the cultivator and ensuring its efficiency, versatility, and effectiveness in agricultural operations. It is used to control the speed and torque of the cultivator, providing consistent power for tasks like soil tilling and field preparation.

Types of Motors Used in Cultivators

There are two primary types of motors commonly used in electric cultivators:

1. BLDC Motor (Brushless DC Motor)

- **Definition:** A BLDC motor is a type of electric motor that uses direct current and a brushless design, which eliminates the need for brushes in the motor's operation. This design results in improved efficiency, durability, and performance compared to traditional brushed motors.
- **Usage:** BLDC motors are widely used in cultivators due to their high efficiency, low maintenance, and ability to provide constant torque over a wide range of speeds. They are typically used to drive the rotavator blades and wheels of the cultivator.

- **Advantages:** BLDC motors are more efficient, produce less heat, have a longer lifespan, and require less maintenance. Their quiet operation makes them ideal for environments that require low noise, such as urban or residential farming areas.

2. PMSM Motor (Permanent Magnet Synchronous Motor)

- **Definition:** PMSM motors are a type of electric motor where the rotor is magnetized by permanent magnets. These motors are typically used for applications requiring higher power and torque at a constant speed.
- **Usage:** PMSM motors are used in cultivators for tasks that require high torque, such as driving the differential and ensuring effective movement and direction control. They are often chosen for their ability to handle heavy loads and provide consistent power output.
- **Advantages:** PMSM motors offer excellent power density, better performance at high speeds, and greater efficiency. They are more suited for heavy-duty applications where torque is a critical factor.

Power Specification Requirement for Cultivators

When selecting a motor for the electric cultivator, it is essential to calculate the power requirements based on several factors:

1. **Torque and Speed Requirements:** The motor must provide sufficient torque to drive the rotavator blades and move the cultivator efficiently across the field. The required speed will depend on the field conditions and the desired operation rate.
2. **Total Load:** The motor must be able to handle the weight of the cultivator, including the frame, motor, battery, and attachments. The motor's power rating should be sufficient to overcome this load.
3. **Terrain Conditions:** If the cultivator operates on uneven or tough terrain, the motor should provide enough torque to ensure consistent performance without overloading.
4. **Operational Efficiency:** Motors with higher efficiency will consume less energy, extending the battery life and reducing operating costs.

4.1.1 Calculating the required torque for the tiller cultivator:

To calculate the required torque for a tiller cultivator, we need to consider several factors related to soil conditions, cultivator design, and operational parameters. The key variables include:

- **Soil Resistance Force:** The resistance that the soil exerts on the tines or blades.
- **Operating Speed (RPM):** The speed at which the tiller will rotate.
- **Tine Radius or Blade Radius (r):** The radius of the rotating tines or blades.
- **Tillage Depth and Width:** The depth and width of the soil being tilled affect the force required.

4.1.2 Soil Resistance force:

The soil resistance (or draft force) of clayey soil depends on various factors, such as moisture content, compaction, and tillage depth.

For clayey soil in typical agricultural conditions, draft force per unit area is often within the range of:

6000 to 8000 N/m² for moderate tillage depths (20–30 cm). (1)

When calculating the draft force (soil resistance) for a rotavator, the draft force per unit area is usually given in units like N/m² (Pascals), which represents the force acting over an area.

To compute the total draft force for the entire tillage width, we need to multiply the draft force per unit area by the area being tilled. Here's how to do it step by step:

Draft Force per Unit Area: This is given in N/m² (e.g., 7000 N/m²).

Calculating Tillage Area:

Width of tillage (in meters): 1 m

Depth of tillage (in meters): 0.25 m

The area being tilled can be considered as:

$$\text{Tillage Area} = \text{Width} \times \text{Depth} = 1 \text{ m} \times 0.25 \text{ m} = 0.25 \text{ m}^2$$

Calculate Total Draft Force:

Total draft force F_{tillage} is given by:

$$F_{\text{tillage}} = \text{Draft force per unit area} \times \text{Tillage Area}$$

For example, if the draft force per unit area is 7000 N/m²:

$$F_{\text{tillage}} = 7000 \text{ N/m}^2 \times 0.25 \text{ m}^2 = 1750 \text{ N}$$

4.1.3 Tillage Depth:

1. Tillage Depth:

The tillage depth refers to how deep the blades penetrate into the soil. It should match the specific agricultural needs and the soil conditions in Nepal.

Below are general recommendations based on soil type and crop needs

Medium Tillage (15–20 cm):

- Common for most row crops like maize, wheat, rice, and barley.
- Provides good soil aeration and moisture retention.
- Suitable for moderately compacted soils (like loamy soils with clay content).

Deep Tillage (20–25 cm):

- Suitable for deep-rooted crops such as sugarcane and certain vegetables like potatoes.
- Helps break up hardpans or compacted layers in clayey soils.

- Useful in heavy clay soils or areas that require subsoil aeration.

Recommendation:

For Nepal's agricultural context (where rice, wheat, maize, and other crops are common), a medium tillage depth of 15 to 20 cm is generally recommended. This depth is efficient for a variety of crops and can handle the average soil resistance.

For most small-scale tillers and farming operations in Nepal,

especially in hilly and terrace regions, a tillage width of 0.8 to 1.2 meters is ideal.

In flat regions like Terai, a wider tillage width of 1.2 to 1.5 meters can be used for larger fields.

Conclusion:

- **Tillage Depth:** 15 to 20 cm is generally suitable for Nepal's varied crops and soil types.
- **Tillage Width:** For small-scale operations, 0.8 to 1.2 meters is recommended, while larger operations in flat regions can go up to 1.5 meters.

These dimensions should work well across different soil types in Nepal, providing a balance between effective tillage and manageable power requirements.

4.1.4 Specification for PMSM motor:

Weight of the Tiller: $W_{\text{tiller}} = 300 \text{ kg}$

Weight of the Rotavator (blades + shaft + casing): 90

Soil Resistance: 1750 N/m^2

Width of the Rotavator: 1 m

Depth of Tillage: 0.2 m

Desired Linear Speed: 1.5 m/s

1. Calculating Draft Force

$$F_{\text{draft}} = \text{soil resistance} \times \text{width of tillage} = 1750 \text{ N/m}^2 \times 1 \text{ m} = 1750 \text{ N}$$

2: Calculating Weight of the Tiller and Rotavator

a. Weight of the Tiller:

$$W_{\text{tiller}} = 300 \text{ kg} \times 9.81 \text{ m/s}^2 = 2943 \text{ N}$$

b. Weight of the Rotavator:

$$W_{\text{rotavator}} = 90 \text{ kg} \times 9.81 \text{ m/s}^2 = 882.9$$

3: Calculating Total Force for PMSM Motor

$$F_{\text{total}} = F_{\text{draft}} + W_{\text{tiller}} + W_{\text{rotavator}} \quad F_{\text{total}}$$

$$= 1750 \text{ N} + 2943 \text{ N} + 882.9 \text{ N}$$

$$= 5585.9 \text{ N}$$

Step 2: Calculate Power Requirement

The power required to maintain the linear motion is calculated using the formula:

$$P = F_{\text{total}} \times v$$

F_{total} is the total force in newtons (N)

- v is the linear speed in meters per second (m/s)

Substituting the values:

$$P = 5575.9 \text{ N} \times 1.5 \text{ m/s} = 8363.85 \text{ W} \approx 8.36 \text{ kW}$$

4.1.5 Specification for BLDC motor:

A Brushless DC (BLDC) motor is a highly efficient, reliable, and durable motor widely used in various applications due to its excellent torque-speed characteristics and minimal maintenance requirements. Its specifications define its performance, mechanical design, and operational parameters, ensuring suitability for specific tasks. Below are the key specifications typically associated with BLDC motors:

Rotavator Blades:

Width of the rotavator = 1 m

Distance between two rotors = 15 cm

$$\text{Number of disks in shaft} = \frac{\text{width of machine}}{\text{distance between two rotors}} = \frac{100}{15} = \geq 6$$

So, number of blades = $2 \times 2 + 4 \times 4 = 20$ blades

(Each disk consists of 4 blades but outermost rotor on both end consists of 2 blades on each)

Arrangement of the blade on the disks at an angular interval A° .

$$A^\circ = \frac{360^\circ}{20} = 18^\circ$$

Calculating the weight of the rotor components:

$$\text{Volume}_{\text{blade}} = \text{Thickness} \times \text{length} \times \text{width} = 0.01 \times 0.25 \times 0.05 = 1.25 \times 10^{-4}$$

$$\text{Total Volume} = \text{Volume}_{\text{blade}} \times 20 = 2.5 \times 10^{-3}$$

Weight = Total x 7850 kg/m³ (Density of mild steel) = 20 kg

Calculating the weight of the shaft:

$$\text{Volume}_{\text{shaft}} = \pi \times \frac{d^2}{4} \times l = 2.83 \times 10^{-3}$$

Weight = Total x 7850 kg/m³ (Density of mild steel) = 22 kg

Known and Estimated Parameters

- **Weight of Rotavator (blades + shaft):** 40 kg
- **Weight of Casing of Rotavator:** 50 kg
- **Total Weight of Rotavator:** 40 kg+50 kg=90 kg
- **Soil Type:** Clayey soil
- **Width of the Rotavator:** 1 m
- **Depth of Tillage:** 20 cm=0.2 m
- **Desired Linear Speed:** 1.5 m/s

Calculate Force Required to Rotate the Rotavator

To rotate the rotavator, we need to determine the torque required.

Let's assume an effective radius for the rotavator blades.

If the blade radius is $r_{\text{rotavator}} = 0.25 \text{ m}$

$$r_{\text{rotavator}} = 0.25 \text{ m}$$

the force can be calculated as:

Torque Calculation:

$$T_{\text{rotavator}} = F_{\text{draft}} \times r_{\text{rotavator}} = 1750 \text{ N} \times 0.25 \text{ m} = 437.5 \text{ Nm}$$

Rotational Speed: Let's use **100 RPM** (which is approximately **10.47 rad/s**) for the rotavator.

Step 3: Calculate Power for the BLDC Motor

Using the torque and rotational speed to find the power required for the BLDC motor:

$$\begin{aligned} P_{\text{BLDC}} &= T_{\text{rotavator}} \times \text{Rotational speed} \\ &= 437.5 \text{ Nm} \times 10.47 \text{ rad/s} \approx 4577.63 \text{ W} \approx 4.58 \text{ kW} \end{aligned}$$

Conclusion: In conclusion, the PMSM motor of 8kw is used to provide the linear motion whereas a BLDC motor of 4.5 kw is used to provide the required rotatory motion to the rotavator.

4.2 Battery:

Li-ion batteries are widely chosen for electric cultivators due to their high energy density, long cycle life, and efficiency in power delivery. These features make them suitable for agricultural applications where the equipment must operate continuously under load for extended periods. Additionally, Li-ion technology supports fast charging, which is valuable for reducing downtime between uses.

Key Advantages of Li-ion Batteries

- **High Energy Density:** Li-ion batteries provide substantial power output in a compact size, allowing for a lighter battery that does not add excessive weight to the cultivator, making it easier to manoeuvre.
- **Long Cycle Life:** With 1000-3000 charge-discharge cycles, Li-ion batteries offer longevity, reducing replacement costs over time and increasing reliability for agricultural use.

- **Efficiency and Low Self-Discharge:** Li-ion batteries have a low self-discharge rate (around 2-3% per month), allowing them to retain charge over periods when not in use.
- **Fast Charging Capability:** Li-ion batteries can be recharged relatively quickly, minimizing downtime. This is especially beneficial for operations requiring long working hours with minimal interruptions.
- Summary

1. Gathering Motor Power Requirements

From previous calculations:

- **PMSM Power Requirement:** 8.36 kW
- **BLDC Power Requirement:** 4.58 kW

2. Determining Total Power Demand

$$P_{\text{total}} = P_{\text{PMSM}} + P_{\text{BLDC}} = 8.36 \text{ kW} + 4.58 \text{ kW} = 12.94 \text{ kW}$$

3: Estimate Operating Time

Let's assume an operating time of **2 hours**

4: Calculate Total Energy Requirement

The energy required from the battery (E) , in watt-hours (Wh) is:

$$E = P_{\text{total}} \times t$$

$$E = 12.94 \text{ kW} \times 2 \text{ h} = 25.88 \text{ kWh}$$

Step 5: Battery Weight Calculation

To estimate the battery weight, we need to know the **specific energy** of the battery (typically measured in Wh/kg). For lithium-ion batteries, a reasonable specific energy density is about **200 Wh/kg**. [2]

Battery Weight Calculation

$$\text{Battery Weight} = \frac{E}{\text{Specific Energy}}$$

$$\text{Battery Weight} = \frac{25880}{200} = 129.4 \text{ kg}$$

For a 2-hour operating time with a total power demand of 12.94 kW, a Li-ion battery with a capacity of 25.88 kWh is required. Given a specific energy of 200 Wh/kg, the estimated battery weight would be **129.4 kg**.

4.3 PTO Shaft:

The PTO shaft is a critical component in the electric cultivator, transmitting mechanical power from the BLDC motor to the rotavator. The shaft allows the rotavator to rotate at the necessary speed and torque to effectively till the soil. By directly linking the motor to the tilling implement, the PTO shaft enables efficient energy transfer, optimising power usage and ensuring consistent rotation under varying soil conditions.

Key roles of the PTO shaft:

- **Power Transmission:** Transfers rotational power from the motor to the rotavator.
- **Durability and Reliability:** Designed to withstand high torques and resist bending or twisting.
- **Energy Efficiency:** Ensures minimal power loss in the drive system, maintaining the cultivator's energy efficiency.

2. Calculations for PTO Shaft Specifications

Motor Power Requirement

Motor Type: BLDC (Brushless DC Motor)

- **Motor Power:** 4.58 kW

This power level ensures sufficient torque and rotational speed for tilling tasks, even in challenging soil conditions.

Rotational Speed (N)

Rotavator Speed: 10.47 rad/s (approximately 100 RPM)

Torque Requirement

- To determine the torque transmitted by the PTO shaft, we use the power-torque relationship:

$$T = \frac{P \times 9550}{N}$$

where:

- T = Torque (Nm)
- P = Power (kW)
- N = Rotational speed (RPM)

Substituting the values:

$$T = \frac{4.58 \times 9550}{100} \approx 437.59 \text{ Nm}$$

Calculated Torque: 437.59 Nm

Shaft Diameter Calculation

To handle this torque, we calculate the required shaft diameter based on the material strength. Assuming the shaft is made of mild steel with an allowable shear stress (τ) of 40 MPa, the diameter calculation is as follows:

Using the torsional strength formula:

$$T = \frac{\pi \times d^3 \times \tau}{16}$$

Rearranged to solve for d^3 :

$$D = \left(\frac{16 \times T}{\pi \times \tau} \right)^{\frac{1}{3}}$$

Substituting values:

- **Torque (T):** 437.59 Nm

- **Shear Stress (τ):** 40 MPa

The calculated diameter is approximately **38.2 mm**.

For practical purposes, a standard shaft diameter of 40 mm is recommended to provide additional durability and account for any unexpected load or wear over time.

Summary of PTO Shaft Specifications

- **Material:** Mild Steel
- **Torque Requirement:** 437.59 Nm
- **Rotational Speed:** 100 RPM (10.47 rad/s)
- **Calculated Diameter:** 38.2 mm (rounded to 40 mm for additional safety)

This design ensures that the PTO shaft can effectively and safely transmit power from the BLDC motor to the rotavator, meeting the mechanical demands of tilling operations.

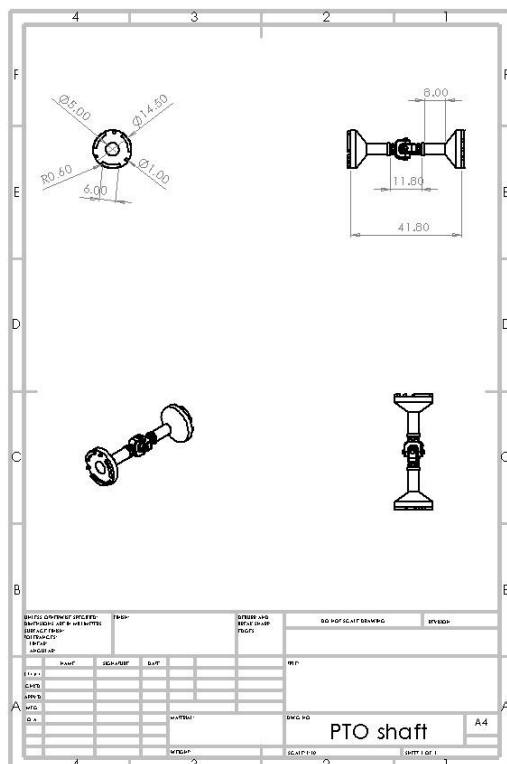


Fig 1 Detailed drawing of PTO shaft

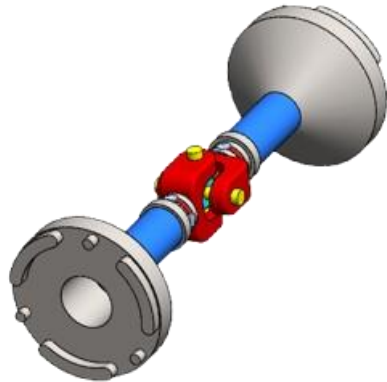


Fig 2 Designed PTO shaft in solidworks

4.4 Differential along with Axle:

In this electric cultivator, a BLDC motor powers the PTO shaft to rotate the rotavator, while a separate 8 kW PMSM motor operates the 42" differential for linear motion, enabling smooth forward and backward movement.

The **PMSM motor** exclusively drives the differential, handling the cultivator's linear motion. With an 8-kW power rating, the PMSM motor supplies enough torque to move the cultivator steadily across various terrains. The differential allows for independent wheel rotation, which is essential for stability and manoeuvrability, especially during turns or when navigating uneven ground.

The **axle** connects the differential to the wheels, transmitting the power required for traction and balance. The axle withstands the linear and torsional loads generated by field operations. Large, wide-tread wheels further enhance stability, offering ground clearance and grip on loose or muddy soils while reducing compaction.

This dual-motor setup, with the BLDC motor dedicated to the rotavator and the PMSM motor managing linear motion, provides precise control, efficiency, and adaptability, making this electric cultivator a robust solution for modern, sustainable agriculture.

The electric cultivator will utilise a 42-inch imported differential, powered by a PMSM motor to provide the necessary torque and power for movement. The differential, serving its core function, will efficiently manage and control the direction of the cultivator, ensuring smooth manoeuvrability. Additionally, the axle of the differential is designed to include attachments that integrate with the steering system, further enhancing the cultivator's handling and operational precision. This configuration ensures reliable performance and adaptability to varying terrains.

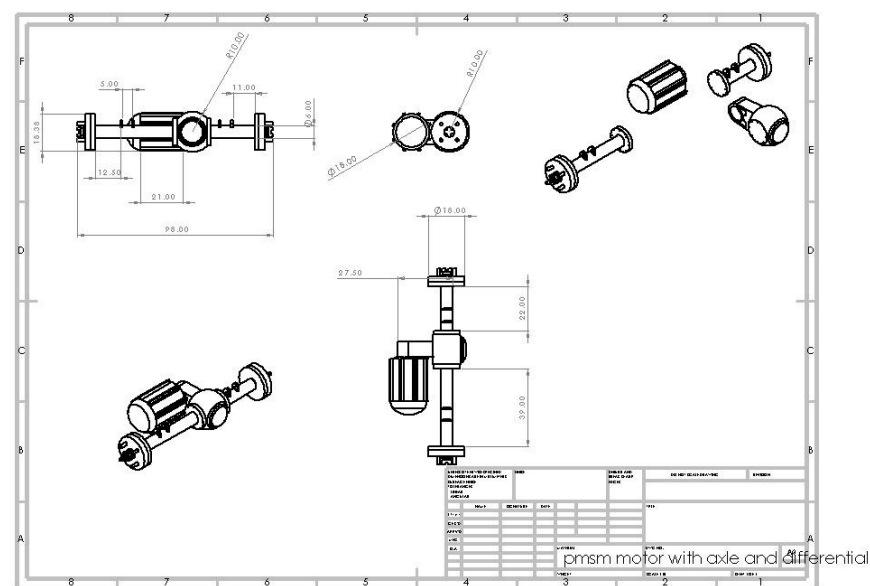


Fig 3 Detailed drawing of the pmsm motor attached with the differential and axle



Fig 4 Designed components of PMSM motor, differential along with axle in solidworks

4.5 Tyres:

Tyres and rims play a crucial role in ensuring the effective operation of a cultivator by providing traction, stability, and support. For agricultural machinery like a cultivator, tyres are designed to perform optimally in varying terrains, such as loose soil, mud, or rugged fields. The rims securely hold the tyres and are engineered to withstand the weight and operational stresses during cultivation.

The selection of tyres and rims depends on factors such as load-carrying capacity, terrain type, and machine weight. For this electric cultivator, standard tyres with a diameter of 50 cm are chosen for their balance of durability, traction, and adaptability to uneven agricultural landscapes.

Specification of standard tyres:

1. **Tyre Type:** Agricultural tread pattern (R1 or R4) for enhanced grip in soil and muddy conditions.
2. **Diameter:** 50 cm (20 inches).
3. **Width:** 12–16 cm, depending on the required contact area and traction.
4. **Rim Diameter:** Compatible with a 16–18 inch rim.
5. **Load Capacity:** Up to 500 kg per tyre, suitable for the cultivator's load.
6. **Ply Rating:** 6–8 plies for added strength and durability.
7. **Material:** High-grade rubber with reinforced sidewalls for resistance to cuts and punctures.
8. **Inflation Pressure:** Typically, 1.5–2.5 bar, ensuring optimal contact and performance.
9. **Tread Depth:** Approximately 20–30 mm for better traction in soft and loose soil.

This configuration ensures that the tyres provide the necessary support and traction for efficient operation, while the rims contribute to the overall structural integrity of the wheel assembly.

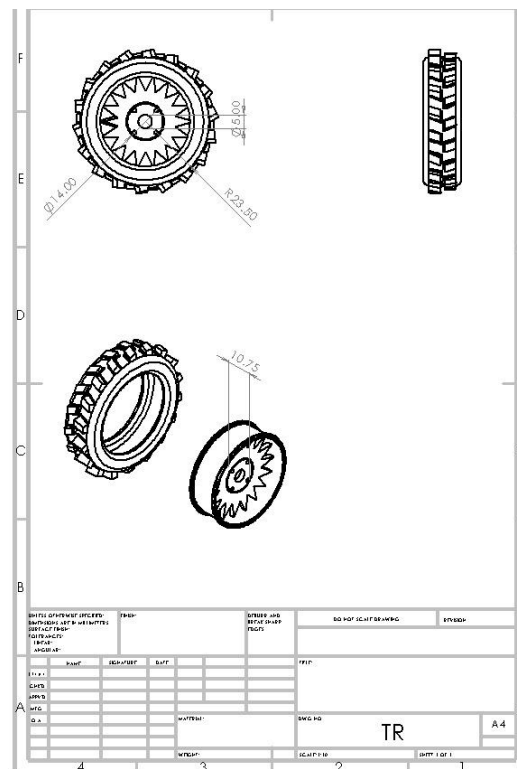


Fig 5 Detailed drawing of tyre and rim

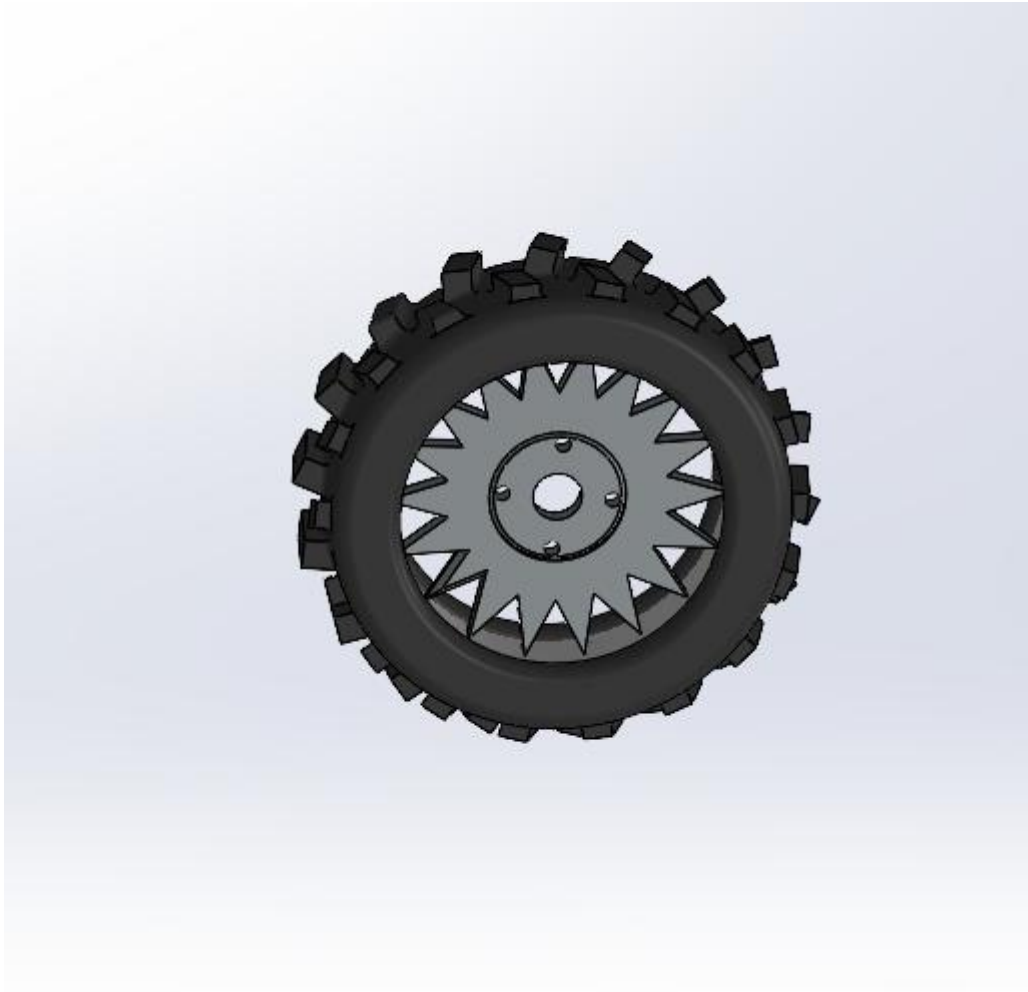


Fig 6 Designed component of tyre from solidworks

4.6 Controller

To control the electric motors in this electric cultivator, an electronic motor controller is essential for regulating speed, torque, direction, and response based on the specific needs of each motor. Since both a BLDC motor and a PMSM motor are used, each requires its own type of controller optimized for its performance characteristics.

1. Controller for the BLDC Motor (Driving the PTO Shaft)

The **BLDC motor controller** manages the rotation of the PTO shaft to power the rotavator. Since BLDC motors operate on a series of precise, electronically-commutated phases, this controller's main functions are:

Electronic Commutation: The controller uses Hall sensors or encoder feedback to detect rotor position and switch the phases accordingly, ensuring smooth and efficient operation.

Speed and Torque Control: Adjusts the current supplied to the motor, allowing the user to control the rotavator's speed based on soil conditions. Higher torque can be applied when needed, such as in dense soil.

Protection Features: Includes overcurrent, overvoltage, and thermal protection to prevent damage under load.

The BLDC controller can be paired with a user interface or a remote control to enable real-time speed adjustments.

2. Controller for the PMSM Motor (Driving the Differential for Linear Motion)

The **PMSM motor controller** handles the cultivator's linear motion through the differential, providing smooth and responsive speed and torque adjustments. This controller typically uses **Field-Oriented Control (FOC)** to optimize the PMSM motor's efficiency and torque output.

Key functions include:

Precise Positioning and Torque Control: FOC enables high-precision control over the motor's magnetic field, providing better response and energy efficiency. This is crucial for maintaining a consistent speed during linear movement.

Real-Time Feedback: Utilizes sensors (typically encoders) to continuously monitor motor position and speed, adjusting current flow to maintain desired motion.

Advanced Protection Mechanisms: Includes fault-detection and protection circuits for overcurrent, overheating, and stall protection to ensure safe operation, especially during start-stop cycles.

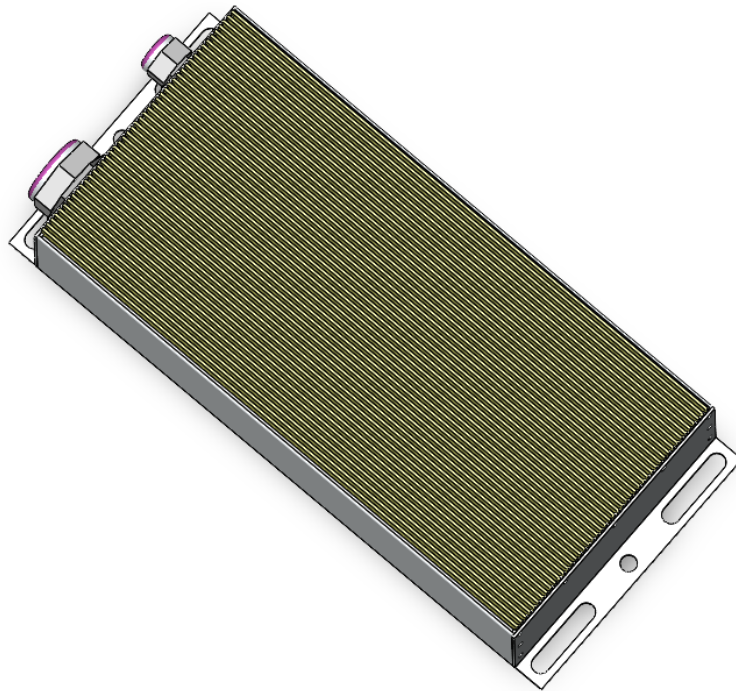


Fig 7 Controller

4.7 Connector between Axle and the Steering:

The connection between the axle and the steering system is established using pins and bolts, ensuring a secure and reliable assembly. This connector serves a dual purpose: it maintains the upright position of the steering system, ensuring stability during operation, and facilitates precise directional control. By transmitting the steering input to the differential, it allows the cultivator to manoeuvre effectively, even in challenging agricultural terrains. The robust design of the connector, with a weight of 2.93 kg, ensures durability and resistance to wear, accommodating the mechanical stresses encountered during field operations.

4.8 Steering System:

The steering system of the electric cultivator is a mechanically operated setup, designed to provide reliable and straightforward control for the operator. It uses conventional steering linkages and components to ensure smooth and precise manoeuvring, particularly in small fields and tight corners. The mechanical nature of the system ensures durability and ease of maintenance, while its simplicity makes it cost-effective and user-friendly. With a total weight of 8.588 kg, the steering system is robust enough to withstand the demands of agricultural operations while remaining manageable for installation and use. This design allows operators to steer the cultivator effectively across different terrains, supporting its role in versatile agricultural applications without the need for complex electronic controls.

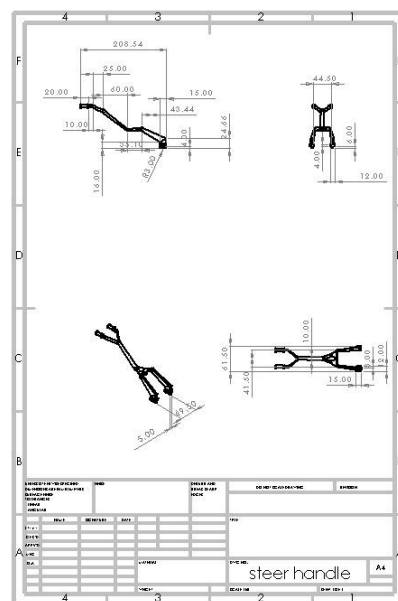


Fig 10 Detailed drawing of the steering

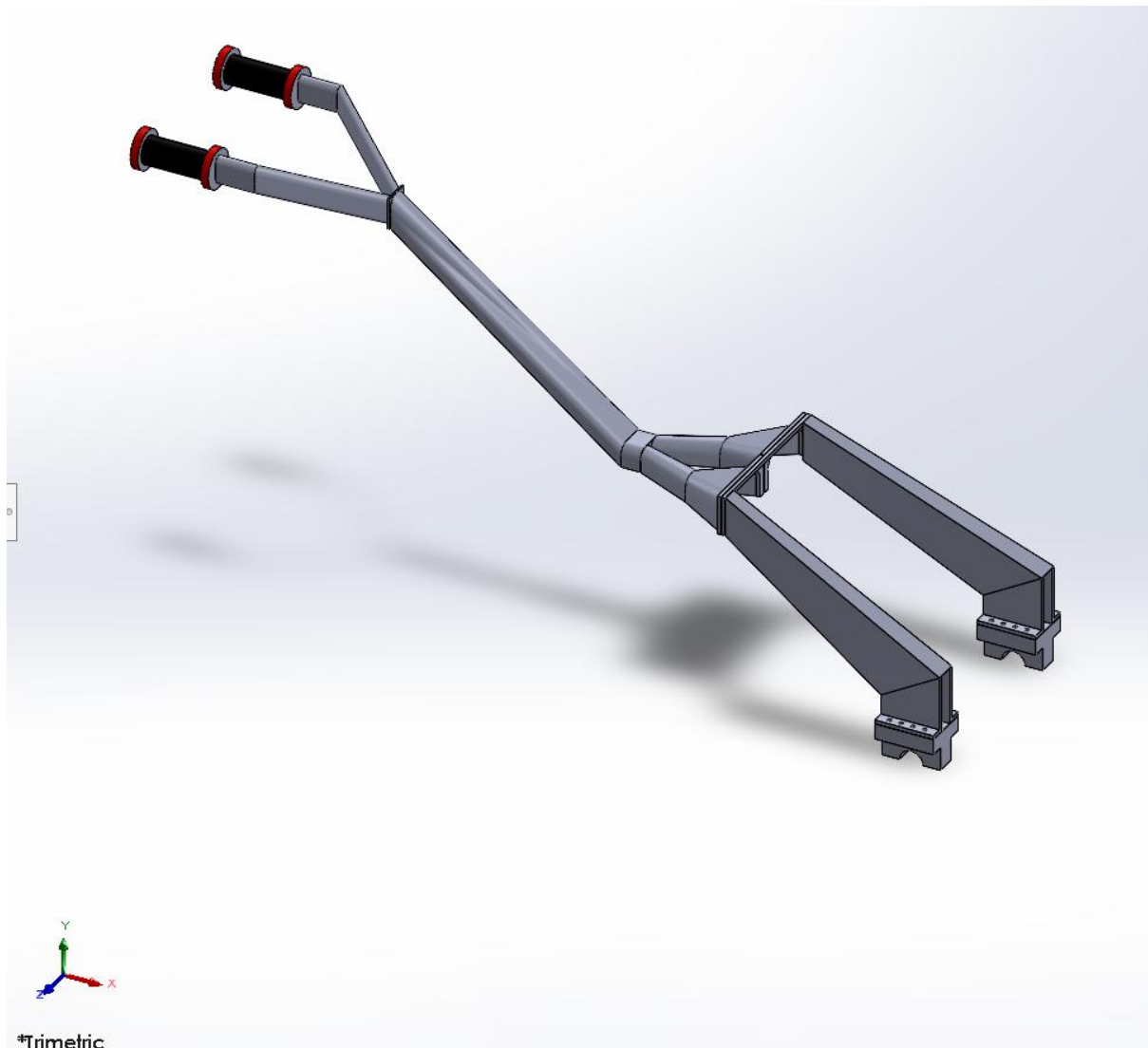


Fig 11 solidwork design of the steering system

4.9 Frame

The frame of the electric cultivator serves as the primary structural component, providing a robust and stable foundation for mounting all other systems, including the motor, rotavator, steering mechanism, and battery. Typically constructed from high-strength materials such as

mild steel or alloy steel, the frame is designed to withstand dynamic loads, vibrations, and stresses encountered during cultivation. Its design ensures optimal weight distribution, enhancing the cultivator's balance and stability while operating on uneven terrain. The frame, with an approximate weight of 100 kg, offers the necessary strength to support all mounted components while maintaining structural integrity. Additionally, it incorporates provisions for attaching implements, ensuring versatility and adaptability for various agricultural tasks.

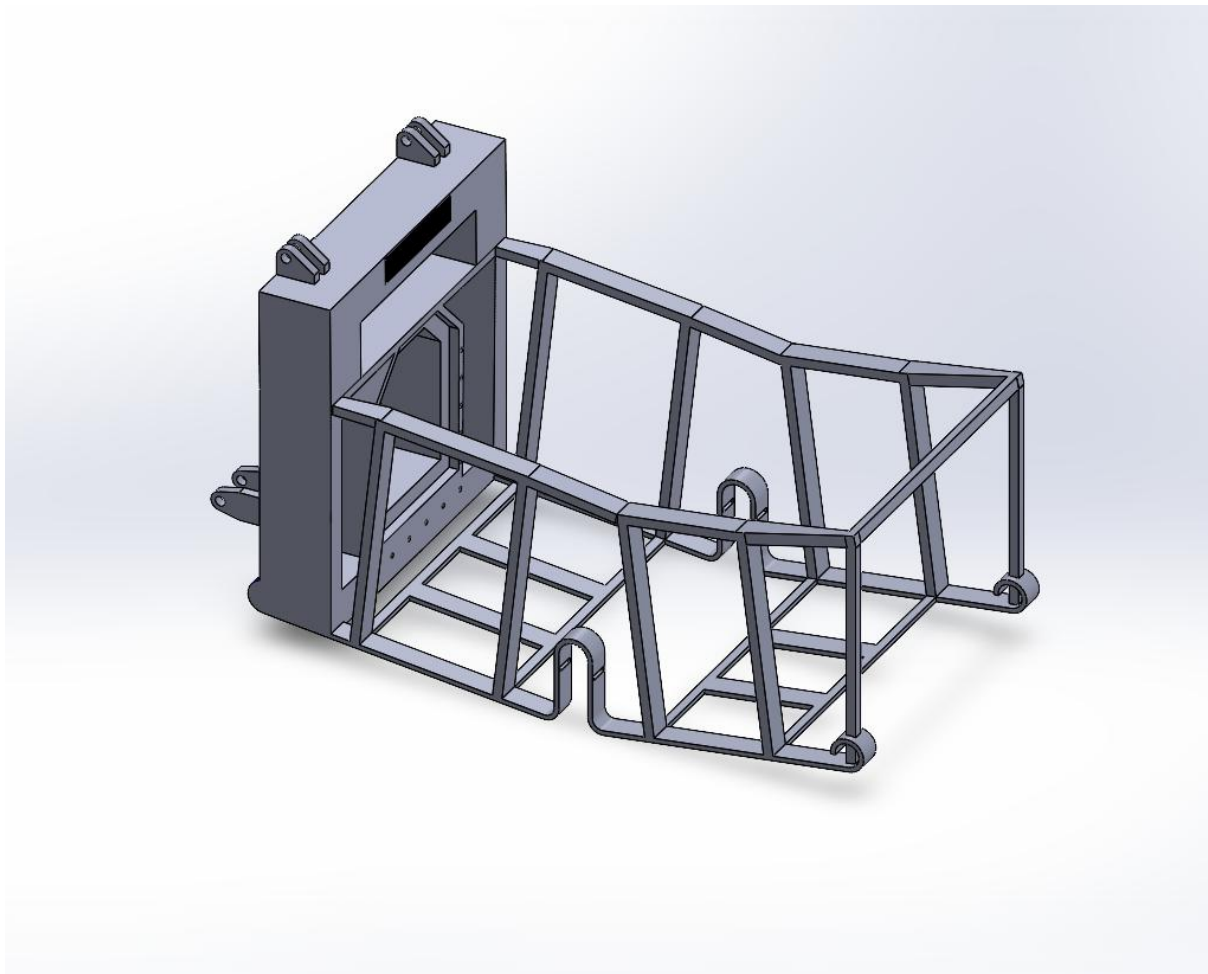


Fig 12 Design of the frame

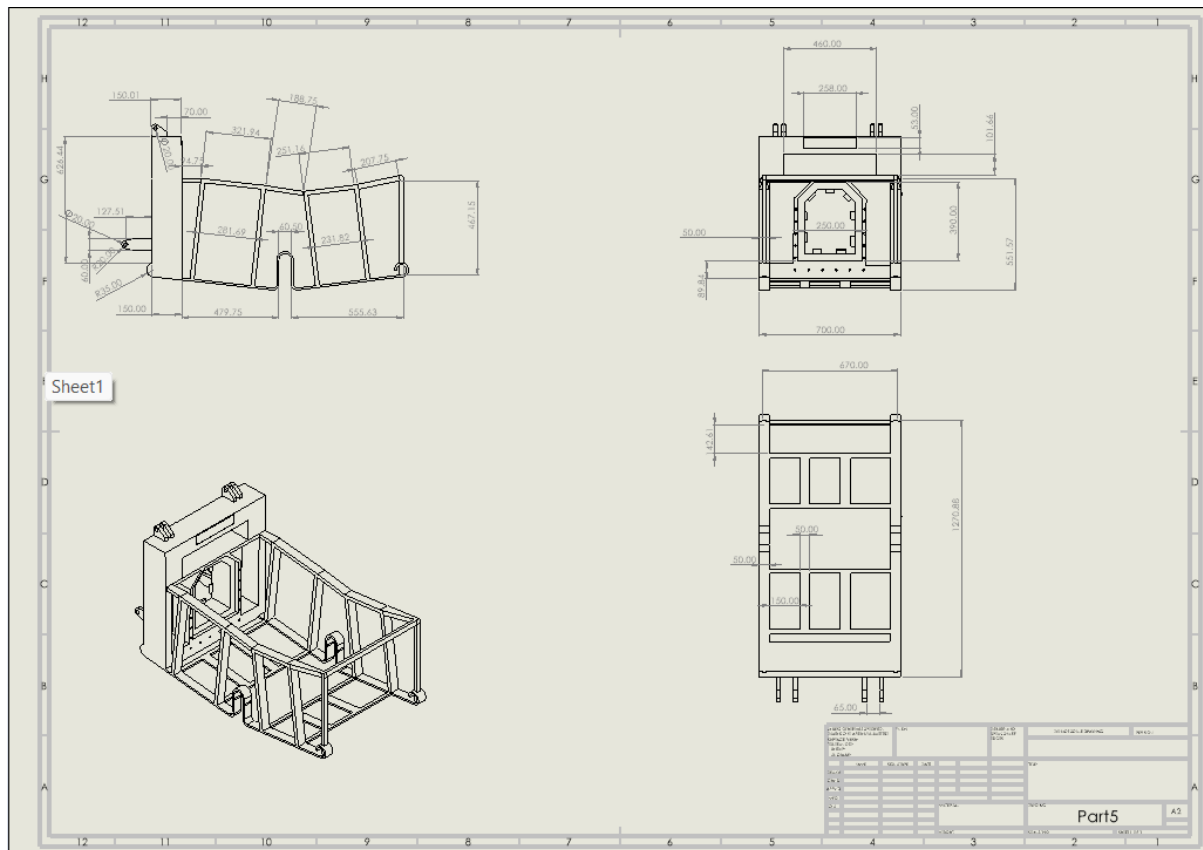


Fig 13 Detailed drawing of the frame

4.10 PTO Gear Box:

The PTO (Power Take-Off) gearbox serves as a crucial component for transferring mechanical power from the BLDC motor to the PTO shaft of the electric cultivator. It is designed to convert the rotational energy generated by the motor into the appropriate torque and speed required for the cultivator's operation. The PTO gearbox is typically mounted between the motor and the PTO shaft, enabling efficient power transmission.

Key Functions and Features of the PTO Gearbox:

1. **Power Transmission:** The primary function of the PTO gearbox is to transmit the rotational power from the motor to the PTO shaft, which drives the rotavator and other attachments of the cultivator.
2. **Speed and Torque Conversion:** The gearbox adjusts the speed and torque of the motor output to match the operational requirements of the cultivator, ensuring optimal performance during use.

3. **Durability and Efficiency:** The gearbox is designed to withstand high loads and stresses, ensuring smooth and efficient power transfer while maintaining durability in harsh agricultural conditions.
4. **Mounting and Integration:** It is securely mounted between the BLDC motor and the PTO shaft, ensuring stable connection and minimal wear over time.
5. **Maintenance:** Regular lubrication and maintenance are necessary to ensure the smooth functioning of the PTO gearbox, reducing the risk of damage and ensuring long-term reliability.

In summary, the PTO gearbox ensures that the power generated by the BLDC motor is efficiently transmitted to the cultivator's working components, providing the necessary movement and functionality for the cultivator to operate effectively.



Fig 14 PTO gearbox

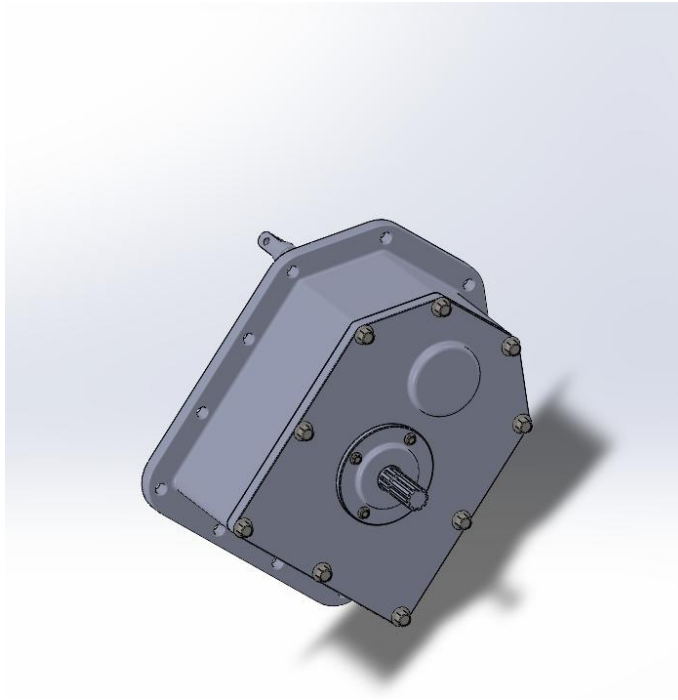


Fig 15 Design of the PTO gear box

4.11 Motor Support:

The support for the BLDC motor is designed to securely mount the motor, ensuring a stable and precise alignment with the PTO transmission box. This support acts as a structural interface, transmitting the power generated by the motor to the PTO transmission box without any misalignment or vibrations that could affect performance. The design ensures that the motor remains securely fixed, even under dynamic loads, allowing for efficient power transfer. Additionally, the support is constructed from durable materials to withstand the operational stresses of the cultivator, ensuring long-term reliability and minimal maintenance.

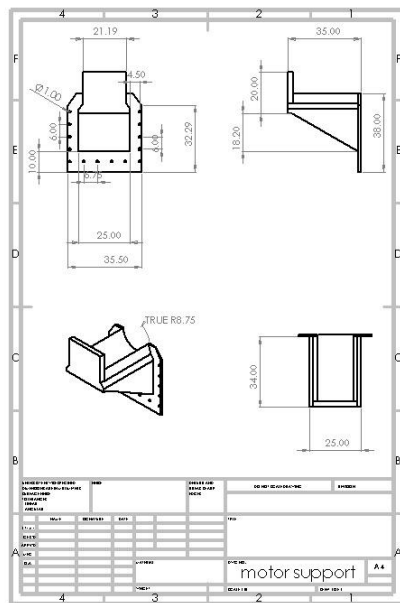


Fig 16 Detailed drawing of the motor support

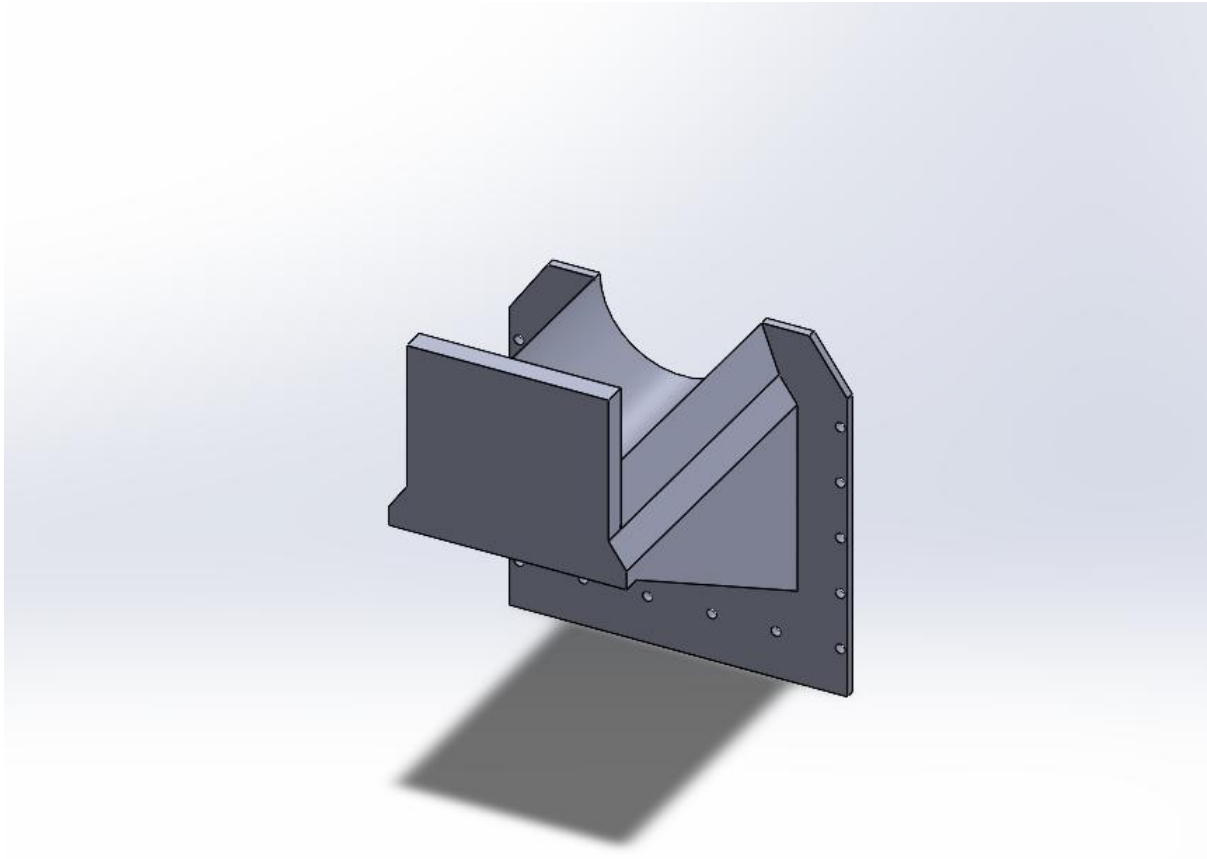


Fig 17 Motor Support 3D design

4.12 Rotavator:

The rotavator is a critical implement of the electric cultivator, designed to efficiently till the soil by breaking clods, aerating the ground, and creating a fine seedbed. It features a horizontal shaft mounted with 22 blades for optimal performance. The blade arrangement includes 2 blades at each end and 4 blades on each of the 5 disks in between, ensuring uniform tillage across the entire working width. The blades, made of medium carbon low alloy steel, are robust and wear-resistant, capable of withstanding the rigours of soil cultivation while maintaining a long service life.

The shaft, crafted from mild steel with a diameter of 40 mm and a length of 1 m, provides the necessary strength and stability to handle dynamic loads during operation. The rotavator is powered by a BLDC motor via a PTO shaft, delivering consistent torque to rotate the blades effectively. With a blade radius of 25 cm, the rotavator achieves a suitable tillage depth, loosening compacted soil layers and promoting better water infiltration and root growth.

The 5 intermediate disks, evenly spaced along the shaft, ensure that the blades cover the soil uniformly, mixing organic matter and fertilisers effectively. This arrangement enhances soil fertility and prepares the field for sowing. The rotavator's design is compact and efficient, making it well-suited for small-scale farming and precise fieldwork, including operations in tight or uneven terrains. By integrating this rotavator into an electric cultivator, farmers benefit from reduced emissions, lower operating costs, and environmentally friendly practices without compromising on tillage quality.

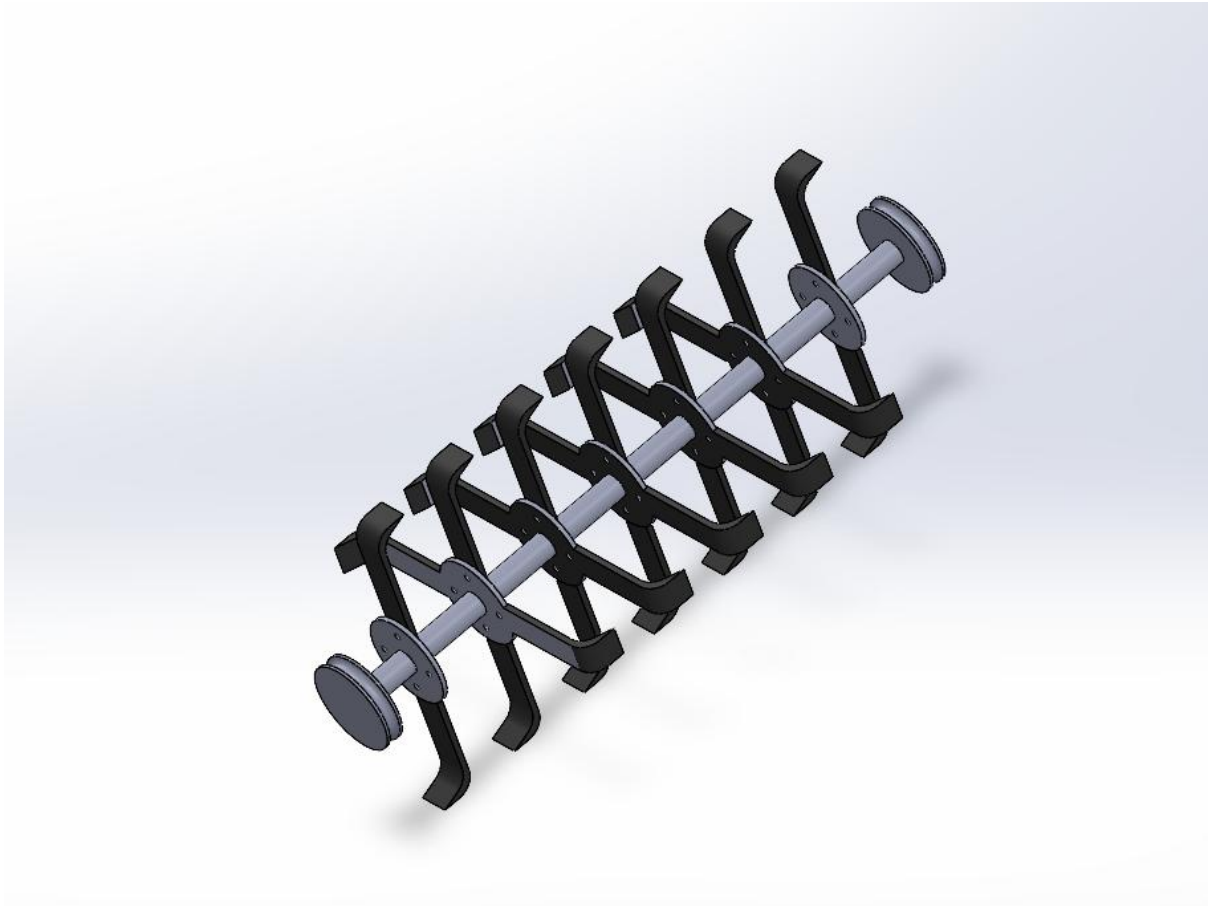
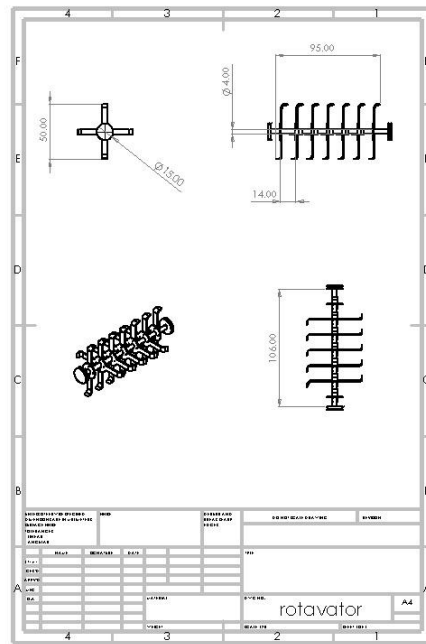


Fig 18 Design of rotavator blades



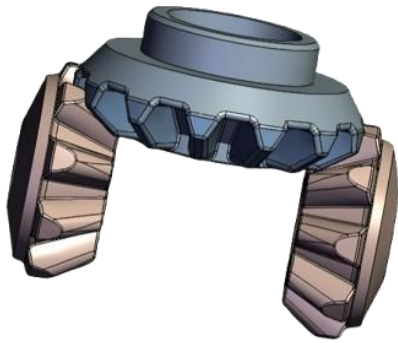


Fig 20 Bevel gears to change the direction of the power supplied by the BLDC motor



Fig 21 : Shaft that will attach with bevel gear and the pulley

On the perpendicular shaft, a pulley system is integrated to further transfer the rotational motion to the horizontal rotavator shaft. The pulley system is selected based on the required speed reduction or amplification, ensuring the rotavator operates at optimal RPMs for effective soil penetration and mixing. This arrangement allows the rotavator blades to rotate uniformly, achieving consistent tillage depth and width.

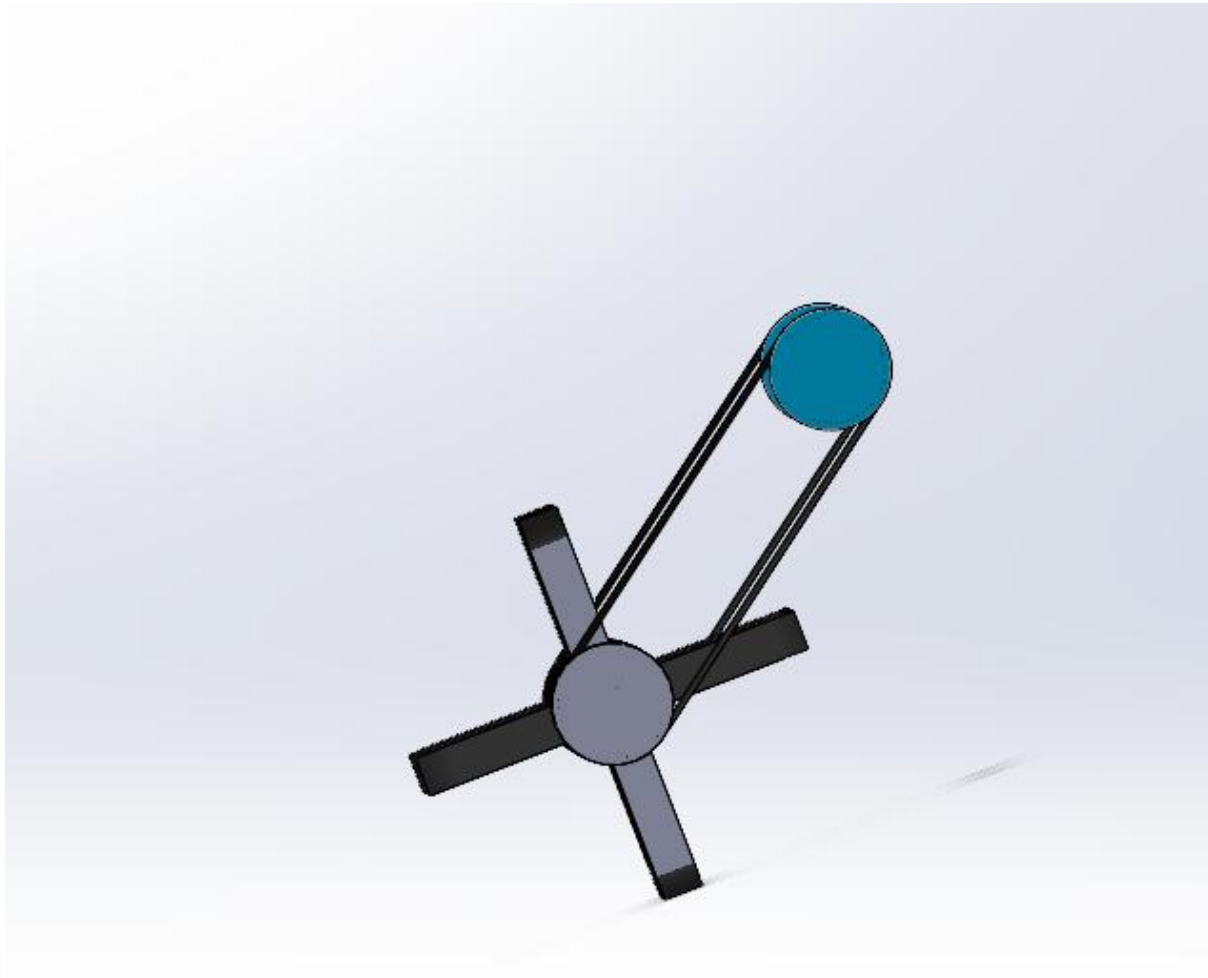


Fig 22 Rotavator driven by chain drive

The assembly's design ensures efficient power transmission with minimal losses, while the use of a pulley system provides flexibility in adjusting the rotavator's speed as per the soil conditions and farming requirements. The robust construction of the PTO shaft and pulleys ensures durability and reliable performance under varying operational stresses. This configuration makes the rotavator assembly versatile, compact, and well-suited for small-scale agricultural applications.

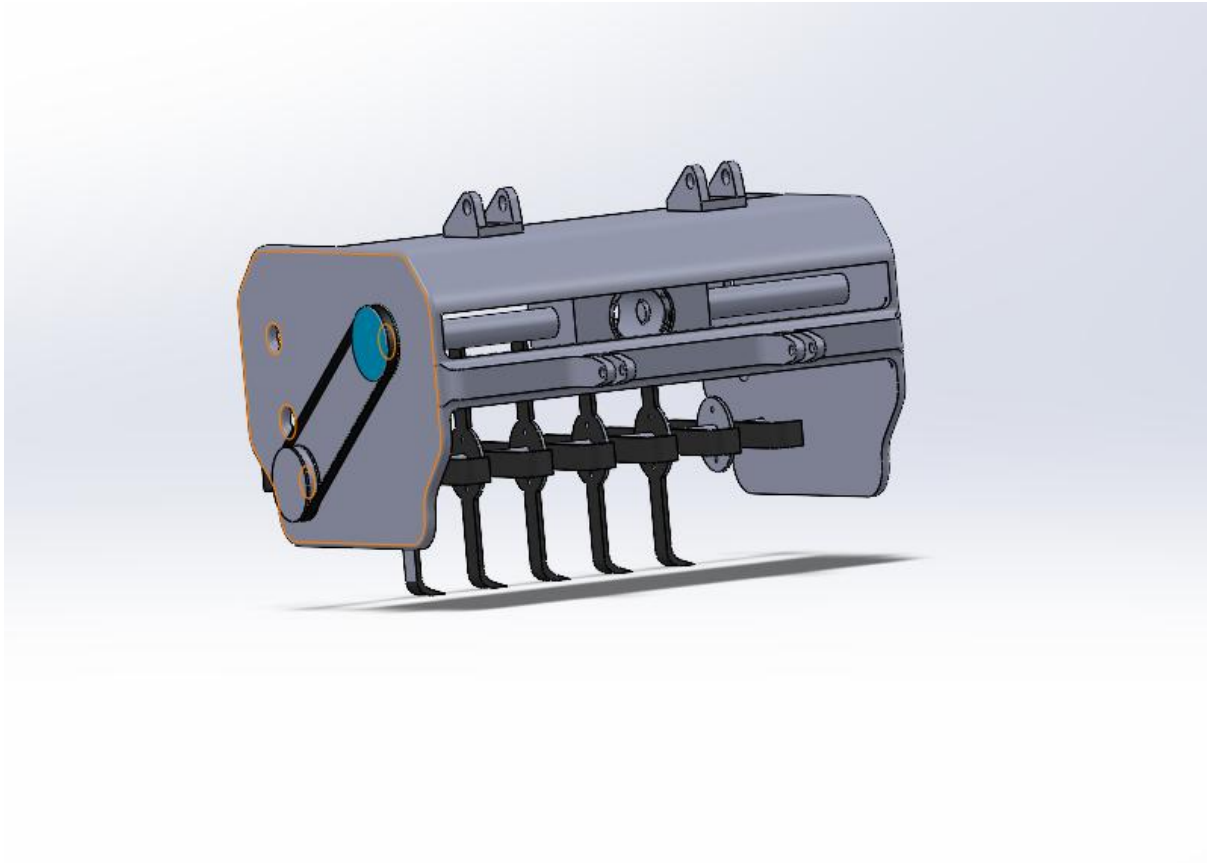


Fig 23 Rotavator assembly

4.14 Detailed Design of the machine:

The designed tiller is an electric-powered cultivator that incorporates multiple mechanisms to perform efficient soil preparation. Each mechanism is carefully designed to optimise power transmission, soil tillage, and operator control. The key mechanisms of the tiller are as follows:

1. Power Transmission Mechanism

The tiller utilises a BLDC motor to generate rotational power, which is transferred to a PTO (Power Take-Off) shaft. The PTO shaft transmits power to a perpendicular shaft through a bevel gear system, ensuring a 90-degree orientation change. From the perpendicular shaft, a pulley system is used to drive the rotavator shaft. This configuration provides flexibility in adjusting the rotavator's speed and ensures consistent torque delivery for effective operation. The use of a pulley allows for speed variation based on the soil type and desired tillage quality.

2. Rotavator Mechanism

The rotavator mechanism is the core tillage component. It consists of a horizontal shaft mounted with 22 blades, with 2 blades at each end and 4 blades on each of the 5 disks in between. The blades are made of medium carbon low alloy steel, chosen for their strength and wear resistance. When powered, the shaft rotates the blades, cutting through the soil to achieve a uniform tillage depth and width. The rotavator mixes the soil, organic matter, and fertilisers effectively, creating an ideal seedbed.

3. Linear Motion Mechanism

The tiller's movement is facilitated by a PMSM motor that powers the differential mechanism. The differential distributes power to the wheels, enabling smooth linear motion of the tiller. This setup allows for precise control over the tiller's forward and reverse motion, aiding in manoeuvrability and efficiency during fieldwork.

4. Steering Mechanism

The steering system of the tiller is a mechanically operated setup that ensures reliable and precise directional control. It employs conventional steering linkages, making it simple, durable, and easy to maintain. This mechanism enables operators to navigate tight corners and uneven terrains effectively.

5. Support Frame Mechanism

The support frame serves as the structural backbone, housing all the components, including the motor, rotavator, steering system, and wheels. Constructed from mild steel or alloy steel, the frame ensures durability, stability, and optimal weight distribution. It also incorporates mounting points for attachments, adding versatility to the tiller.

6. Control and Power Supply System

The tiller is powered by a Li-ion battery, which supplies electricity to the BLDC and PMSM motors. The control system includes throttle controls for adjusting the motor speeds and switches for operational functions. This system allows the operator to regulate the tiller's speed, power delivery, and manoeuvrability with ease.

Summary

The designed tiller integrates advanced mechanisms for power transmission, soil tillage, motion, and control. Its efficient configuration minimises power losses and enhances productivity while being environmentally friendly and suitable for small-scale farming operations.

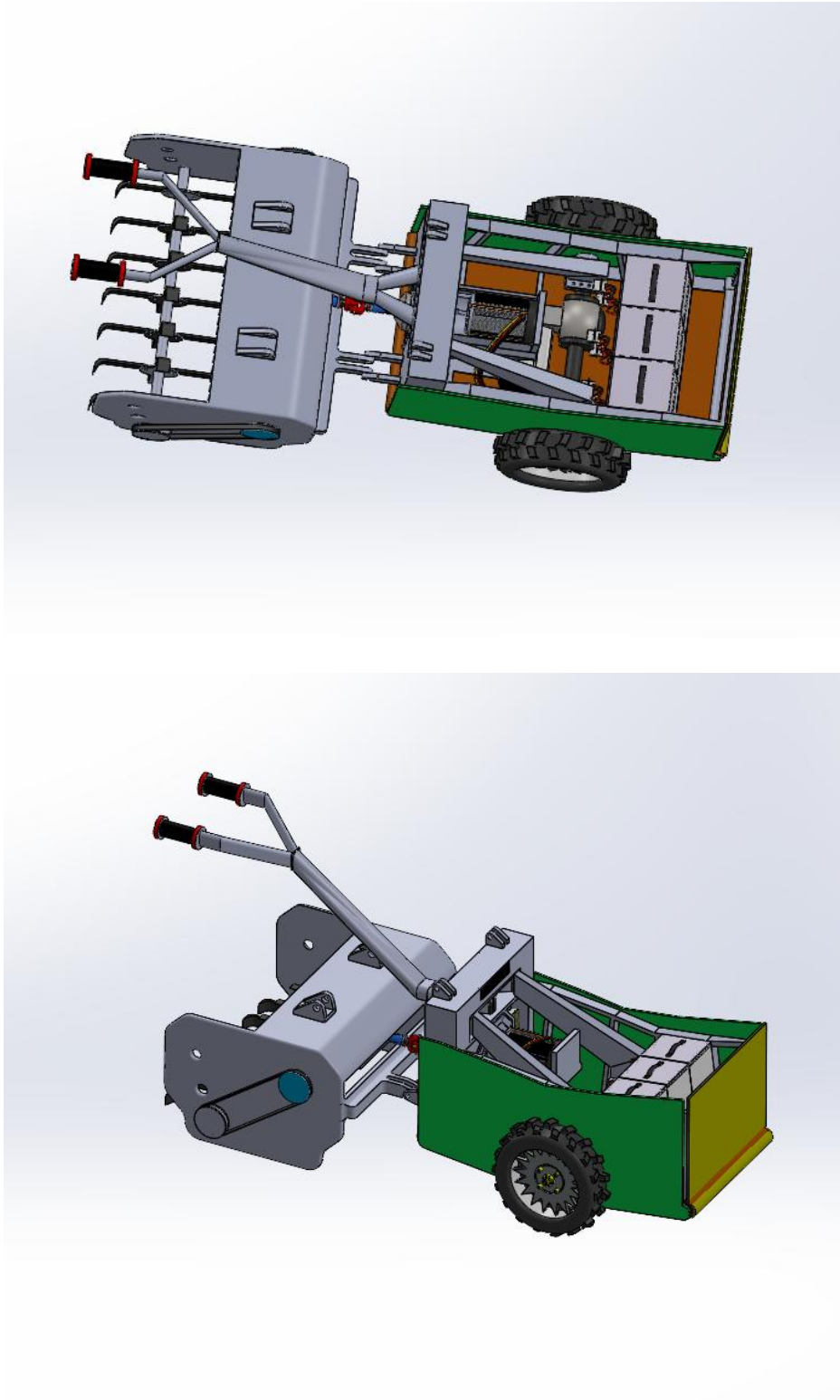


Fig 24 Assembled design of the rotavator

Chapter 5 Electrical Operation of the Cultivator:

The power flow and operational mechanism of the electric cultivator involve several key stages, from drawing power from the battery to executing precise, controlled movements for tillage. Here's an overview of the power flow and how each component works together to enable efficient operation.

1. Power Flow from the Battery

The process begins with the lithium-ion (Li-ion) battery, which serves as the main power source for all electronic components. The battery supplies direct current (DC) power to both the motor controllers and auxiliary systems. Here's a breakdown of the battery's role:

- **Battery Output:** The Li-ion battery outputs DC power, which is managed by the Battery Management System (BMS) to ensure stable voltage and current. The BMS regulates charge levels, monitors temperature, and protects against overcharge or discharge.
 - **Power Distribution:** The power from the battery is directed primarily to two motor controllers—one for the BLDC motor and another for the PMSM motor. These controllers then convert the DC power into suitable forms of current for their respective motors.
-

2. Motor Controllers and Power Conversion

Each motor has a dedicated controller that plays a crucial role in power conversion and delivery.

BLDC Motor Controller (Driving the Rotavator through the PTO Shaft)

- **DC to AC Conversion:** The BLDC motor controller converts DC power from the battery into three-phase AC power. This AC current is required to drive the brushless DC motor and is controlled through electronic commutation.
- **Electronic Commutation:** Using Hall sensors or similar feedback, the controller switches phases to the BLDC motor at specific intervals, ensuring efficient and consistent rotation of the rotavator. By adjusting current flow, the controller can vary the rotavator's speed and torque as needed.

PMSM Motor Controller (Driving the Differential for Linear Motion)

- **Field-Oriented Control (FOC):** The PMSM motor controller uses FOC to manage the PMSM motor's speed and torque with precision. This control method adjusts the motor's magnetic fields to provide smooth torque at various speeds, crucial for stable movement across the field.
- **Power Flow Regulation:** The PMSM controller manages the power flow to the motor based on the desired linear speed. This allows the cultivator to maintain consistent motion, regardless of changing field conditions.

3. Power Transmission to Mechanical Components

Once power reaches the motors, it's converted into rotational or linear motion for specific functions.

PTO Shaft and Rotavator Operation (BLDC Motor)

- **PTO Shaft Drive:** The BLDC motor drives the PTO shaft, transmitting rotational power directly to the rotavator. This shaft acts as a mechanical link, ensuring that the rotational force from the BLDC motor is transferred efficiently to the rotavator blades.
- **Rotavator Motion:** The rotavator blades are rotated by the PTO shaft, creating the necessary tilling action to break up the soil. The speed and depth of the tillage can be adjusted through the BLDC motor controller, providing flexibility for different soil types.

Differential and Axle Drive (PMSM Motor)

- **Differential Control:** The PMSM motor powers the 42" differential, which distributes the torque to each wheel while allowing them to rotate independently. This function is essential for maintaining balance and ensuring that the cultivator can make turns smoothly.
 - **Axle Transmission:** Through the differential, power is transmitted to the axle, which drives the wheels of the cultivator. The linear speed is controlled by the PMSM motor, enabling the cultivator to move forward or backward with ease.
-

4. Control System and Feedback Loop

The control system serves as the operational interface, allowing the operator to adjust the cultivator's functions in real-time. It relies on feedback from sensors to maintain precision in the power flow.

- **Real-Time Adjustments:** Operators can use the control interface to adjust rotavator speed, set desired linear speed, and monitor battery and motor conditions.
 - **Feedback Integration:** Sensors provide feedback on motor speed, temperature, and position, allowing the controllers to adjust current levels for stable and efficient operation.
 - **Safety and Monitoring:** The control system constantly monitors for faults, overheating, or excessive power draw, ensuring that power flow remains within safe limits.
-

Summary of Power Flow and Operation

The electric cultivator's operational mechanism relies on a smooth flow of power from the Li-ion battery to each motor, which is carefully managed by dedicated controllers. The BLDC motor powers the rotavator's rotational movement through the PTO shaft, while the PMSM motor drives the differential for linear motion. The control system ties everything together, allowing precise control over speed, torque, and direction for efficient tillage across diverse field conditions.

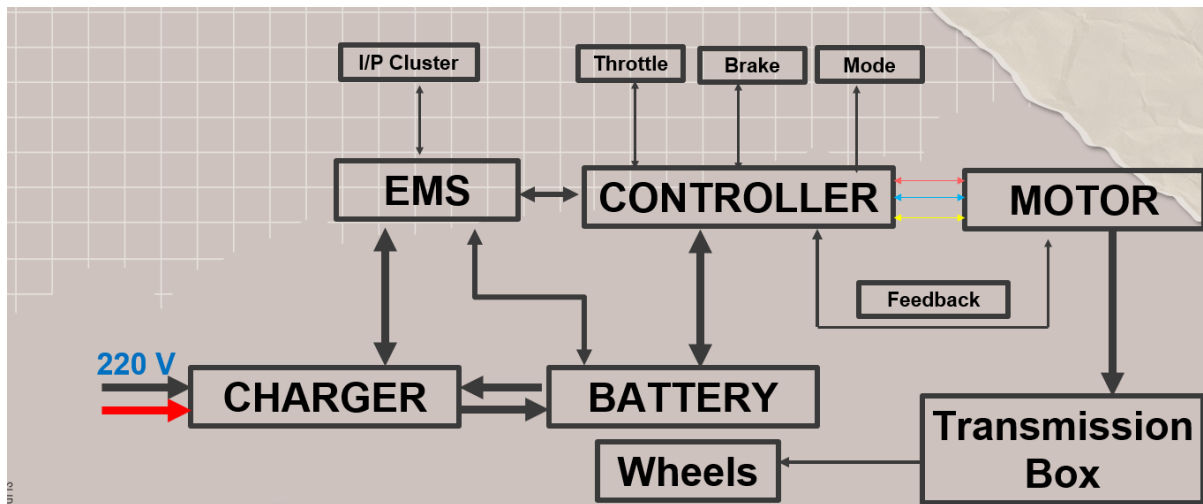


Fig 25 Block diagram of the Electric Vehicle operation

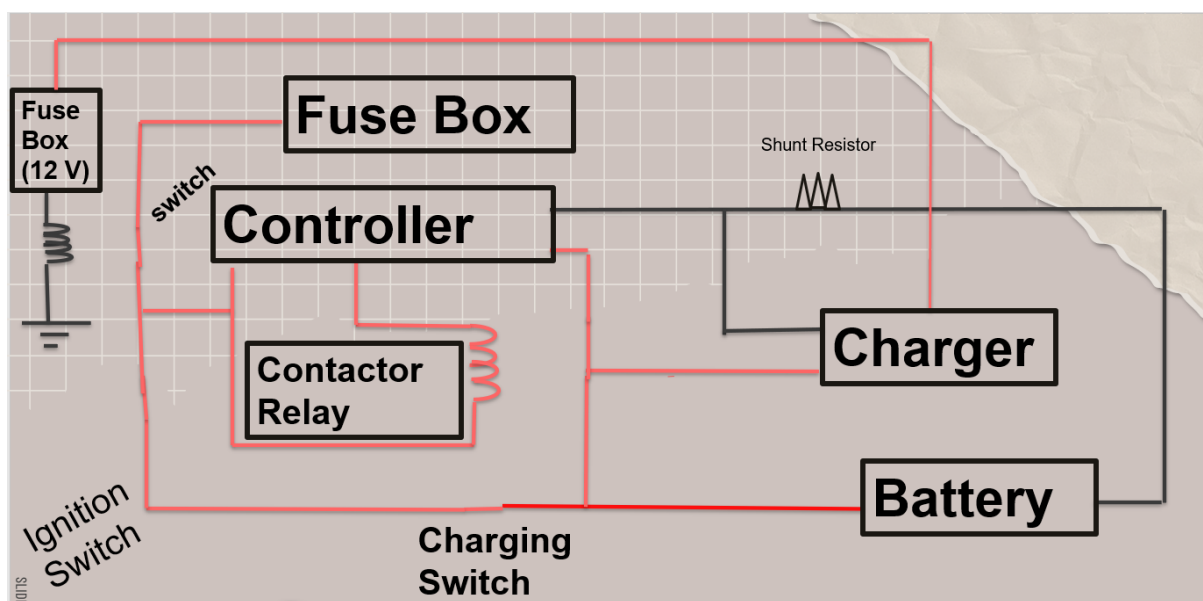


Fig 26 Circuit diagram for the operation of the controller

Chapter 6: Mechanical Operation of the cultivator:

The mechanical operation of an electric cultivator is a well-coordinated interplay of various components that work together to provide efficient soil tillage. The cultivator consists of a

frame, wheels, the rotavator mechanism, and the power transmission system, all designed to facilitate effective soil preparation. Here, we will explore the mechanical operation, detailing how these components function to achieve the desired tillage.

1. Frame and Structure

The **frame** of the electric cultivator serves as the backbone, providing structural integrity and support for all components. Typically constructed from robust materials like mild steel, the frame is designed to withstand the stresses of agricultural operations. It supports the weight of the motors, battery, and other mechanical components while also allowing for easy manoeuvrability across different terrains.

- **Design:** The frame is engineered to keep the centre of gravity low, which enhances stability during operation. Its design often includes adjustable components, allowing the operator to modify the depth of tillage based on soil conditions.

2. Wheels and Axle Assembly

The **wheels** of the cultivator are essential for mobility. They are attached to an **axle** system powered by the **PMSM motor** through a differential mechanism.

- **Differential Mechanism:** The differential allows the wheels to rotate at different speeds while maintaining power transfer from the motor. This feature is particularly important during turns, where the outer wheel needs to rotate faster than the inner wheel to prevent dragging.
- **Wheel Configuration:** The cultivator typically features large, robust wheels designed for traction in various soil types. The design may include tread patterns that improve grip and prevent slippage during operation.

3. Rotavator Mechanism

The **rotavator** is the primary tilling component of the cultivator, responsible for breaking up soil, mixing in organic material, and creating a fine seedbed. It consists of several rotating blades or tines mounted on a shaft, driven by the **BLDC motor** through a Power Take-Off (PTO) shaft.

- **Blade Design:** The blades are typically designed with specific geometries to optimize soil penetration and mixing. They may be curved or straight, depending on the tilling requirements and soil conditions. For instance, curved blades can effectively lift and aerate the soil, while straight blades can chop through tougher material.

Rotation Speed and Torque: The BLDC motor powers the rotavator, allowing for adjustable rotational speeds. This flexibility enables the operator to adapt to varying soil types and conditions, ensuring effective tillage without damaging the soil structure.

4. Power Transmission System

The power transmission system encompasses the mechanisms through which power flows from the electric motors to the mechanical components, allowing for efficient operation.

- **Power Take-Off (PTO) Shaft:** The PTO shaft connects the BLDC motor to the rotavator. It transmits rotational power directly, allowing the rotavator to operate independently of the wheels. This arrangement is critical for situations where the cultivator must move at a different speed than the tilling mechanism operates.
- **Differential Drive:** The PMSM motor powers the wheels through a differential, ensuring that the power is distributed evenly and allowing for smooth operation across uneven terrain.

5. Control and Feedback Mechanism

The mechanical operation of the cultivator is enhanced by its control system, which allows the operator to manage various parameters effectively.

- **User Interface:** The operator can adjust the speed of the rotavator and the cultivator's forward motion using controls integrated into the system. This may include levers, knobs, or touchscreen interfaces that provide real-time feedback on operational status.
- **Sensors:** The cultivator is equipped with sensors that monitor various parameters, including motor speed, torque, and soil conditions. This feedback is used to make real-time adjustments to the motor outputs, ensuring optimal performance.

6. Operational Workflow

When the electric cultivator is powered on, the operator can engage the motors via the control system. The process begins as follows:

1. **Engaging the Motors:** The operator activates the PMSM motor, which drives the wheels, moving the cultivator forward.
2. **Activating the Rotavator:** Simultaneously, the BLDC motor is engaged to rotate the rotavator blades. The operator can adjust the rotational speed based on the required tilling depth and soil conditions.
3. **Tilling Process:** As the cultivator moves across the field, the rotavator blades penetrate the soil, breaking it up and mixing in any organic material. The differential ensures that the wheels maintain traction and stability, even in challenging conditions.
4. **Adjusting Depth and Speed:** Throughout the operation, the operator can monitor soil conditions and adjust the depth of tillage or speed as needed, ensuring efficient and effective soil preparation.

Conclusion

The mechanical operation of an electric cultivator is a complex, synchronized system designed to optimize soil preparation. With a sturdy frame, robust wheels, an efficient rotavator mechanism, and an effective power transmission system, the cultivator is well-equipped to handle the demands of modern agriculture. By leveraging electric motors and advanced control systems, it offers a sustainable and efficient solution for soil cultivation, paving the way for healthier crops and better yields.

Chapter 7: Budget of the Project:

This budget report outlines the financial aspects of the electric cultivator project, including the pricing of components, cost analysis for using a petrol cultivator versus an electric cultivator, and the overall project cost breakdown. This report aims to provide a clear understanding of the project's budget for potential vendors and stakeholders involved in the procurement and fabrication process.

7.1 Pricing Table for Key Components

Below is the pricing table for the essential components of the electric cultivator, providing an overview of the costs involved in assembling the machinery:

Table 1 Cost of the different components of the cultivator

Component	Specification	Estimated Cost (NPR)
Li-ion Battery	Voltage, capacity (e.g., 48V, 50Ah)	1,00,000 – 1,50,000
BLDC Motor	Rotavator operation	15,000–25,000
PMSM Motor	Differential operation (8 kW)	20,000–30,000
Controller	For motor speed and torque control	8,000–15,000
Rotavator Shaft	Mild steel, diameter 40 mm, length 1 m	3,000–5,000
Rotavator Blades	22 blades, medium carbon low alloy steel	10,000–15,000
Frame and Body	Mild steel or aluminium, powder-coated	10,000–20,000
Wheels and Suspension	Standard or customised	5,000–8,000
Electrical Wiring	Cables, connectors, safety fuses	3,000–5,000
Miscellaneous	Bearings, fasteners, covers, etc.	5,000–8,000

Total material cost is 2,00,000 – 3,00,000

2. Fabrication and Assembly

- Machining and Welding: *NPR 10,000–15,000.*
- Surface Treatment: *NPR 5,000–8,000 (coating for corrosion resistance).*
- Testing and Calibration: *NPR 5,000–8,000.*
- Labour Costs: *NPR 15,000–25,000 (depending on complexity).*

Total Fabrication and Assembly Cost: *NPR 35,000–50,000*

7.2 Insights into government policies and subsidy opportunities in Nepal:

1.Mechanisation Support Programs

The Nepalese government actively promotes agricultural mechanisation through subsidies and financial support, particularly for sustainable and energy-efficient technologies:

Subsidy Range: 25–50% of the machinery cost, depending on the region and type of equipment.

Targeted Beneficiaries: Smallholder farmers, cooperatives, and women-led farming enterprises.

2. Green Energy and Sustainability Initiatives

Projects under the **Renewable Energy Subsidy Policy** provide additional incentives for electric-powered agricultural tools.

Collaborations with NGOs: NGOs like the International Fund for Agricultural Development (IFAD) and local organisations often fund sustainable farming projects.

3. Financing and Credit Schemes

Agricultural Development Bank: Provides low-interest loans for purchasing farming machinery.

Cooperatives and Microfinance Institutions: Offer group financing options for small-scale farmers.

7.3 Cost-Benefit Analysis for Farmers in Nepal

To provide a clear comparison between the operational costs of a petrol cultivator and an electric cultivator, the following cost analysis outlines key factors such as fuel/electricity consumption, maintenance, and overall operational costs.

1. Comparison of Initial Investment

While the initial cost is slightly higher, subsidies make the electric cultivator affordable:

Aspect	Electric Cultivator (NPR)	Petrol/Diesel Cultivator (NPR)
Initial Cost	150,000–250,000	120,000–200,000
Subsidy (25–50%)	37,500–125,000	30,000–100,000
Net Farmer Cost	75,000–175,000	90,000–170,000

Table 2 initial price comparison between electric and petrol cultivator

2. Operating Cost Analysis:

The operating cost analysis highlights the significant difference in fuel and maintenance expenses between a petrol and an electric cultivator. While the electric cultivator has a higher initial purchase cost, its operating costs are notably lower. The electricity consumption is more affordable compared to petrol, and the maintenance costs are reduced due to fewer moving parts and the absence of an internal combustion engine. Over time, this results in substantial

savings, making the electric cultivator a more cost-efficient option despite the higher upfront investment.

Parameter	Electric Cultivator	Petrol/Diesel Cultivator
Fuel/Battery Cost	NPR 5–10 per hour (electricity)	NPR 150–200 per hour (fuel)
Maintenance Cost	NPR 1,000–2,000 annually	NPR 10,000–15,000 annually
Total Annual Cost	NPR 3,000–6,000	NPR 25,000–40,000

Table 3 price comparison during the operation of the cultivator

Farmers can save about Rs, 20,000 to 30,000 annually on operational cost

3. Life-time Cost Analysis:

The lifetime cost analysis compares the total expenses of operating a petrol cultivator versus an electric cultivator over a typical service life of 10 years. Although the electric cultivator has a higher initial purchase cost, it offers lower operational and maintenance expenses. The electric cultivator’s energy consumption is significantly cheaper than petrol, and its maintenance costs are lower due to fewer moving parts. Over five years, the electric cultivator’s total cost, factoring in both operational savings and initial investment, is higher initially but results in long-term financial benefits due to reduced fuel and maintenance requirements. This makes the electric cultivator a more cost-effective option in the long run.

Aspect	Electric Cultivator (NPR)	Petrol/Diesel Cultivator (NPR)
Initial Cost	175,000	170,000
Operating Costs	50,000–60,000	250,000–400,000
Maintenance Costs	20,000	100,000–150,000
Total Cost	245,000–255,000	520,000–720,000

. Table 4 Lifetime cost comparison between electric cultivator and diesel cultivator

Over 10 years, the electric cultivator can save farmers about Rs 2,50,000 to 4,50,000.

4. Environmental and Social Benefits

- **Reduction in Fuel Dependency:** Saves approx. 500–1,000 litres of fuel over its lifetime.
- **Lower Carbon Emissions:** Contributes to environmental sustainability, aligning with Nepal’s green energy goals.
- **Ease of Use:** Electric models are quieter and require less mechanical expertise.

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